

Convection Heat Transfer

Syllabus :

Forced Convection : Empirical relations for pipe and tube flow, flow across cylinders, spheres, tube banks. Free convection : Free convection from a vertical, inclined and horizontal surface, cylinder and sphere.

6.1 Forced Convection Correlations : Flow through Circular Pipe

When law of conservation of mass is applied to control volume of flowing fluid, we get **continuity equation** as :

$$\frac{\partial}{\partial x}(\rho \cdot V_x) + \frac{\partial}{\partial y}(\rho \cdot V_y) + \frac{\partial}{\partial z}(\rho \cdot V_z) = 0$$

Application of the **law of conservation of momentum** (i.e. using Newton's second law of motion) gives as :

$$V_x \cdot \frac{\partial V}{\partial x} + V_y \cdot \frac{\partial V}{\partial y} = V \cdot \frac{\partial^2 V}{\partial y^2}$$

(Navier Stokes equation)

The **application of first law of thermodynamics** to the flow results into following **two dimensional heat flow equation** in Cartesian and cylindrical co-ordinates as :

$$k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) = \rho \cdot C_p \left[V_x \cdot \frac{\partial T}{\partial x} + V_y \cdot \frac{\partial T}{\partial y} \right] \quad \dots(6.1.1)$$

In cylindrical co-ordinates.

$$\frac{1}{r} \cdot \frac{\partial}{\partial x} \left(r \cdot \frac{\partial T}{\partial r} \right) + \frac{\partial^2 T}{\partial x^2} = \frac{V}{\alpha} \cdot \left(\frac{\partial T}{\partial x} \right) \quad \dots(6.1.2)$$

The solution of above Equations (6.1.1) and (6.1.2) gives the following co-relations for forced convection in circular pipes.

1. For laminar Flow ($Re < 2000$) :

$$Nu = 4.36 \quad (\text{for constant heat flux}) \dots(6.1.3)$$

$$Nu = 3.66 \quad (\text{for constant wall temperature})$$

$\dots(6.1.4)$

2. For turbulent flow ($Re > 4000$)

$$Nu = 0.023 Re^{0.8} Pr^{0.4} \quad (\text{When fluid is heated i.e. } T_{\infty} < T_w) \quad \dots(6.1.5)$$

$$Nu = 0.023 Re^{0.8} Pr^{0.3} \quad (\text{When fluid is cooled i.e. } T_{\infty} > T_w) \quad \dots(6.1.6)$$

Above equations are known as **Dittus-Boelter co-relations**.

3. Characteristic length of circular pipe, or hydraulic diameter, D_h :

It is possible to predict the flow whether it is laminar or turbulent if we know, ρ , V_{average} , μ and characteristic dimension Le . This dimension Le for flow over the plate is the distance 'x' from leading edge of plate for flow inside tubes of various shapes, its characteristic length is the **hydraulic diameter** defined as

$$D_h = \frac{4 \times \text{Cross-sectional area of flow, } A_c}{\text{Wetted perimeter, } P} \quad \dots(6.1.7)$$

(a) For circular pipe :

$$D_h = 4 \times \frac{\pi D^2}{4} \times \frac{1}{\pi D} = D \quad \dots(6.1.8)$$

(b) For rectangular cross-section conduit of length, L and width, W

(Refer Fig. 6.11.1)

$$D_h = \frac{4A}{P} = \frac{4(L \times W)}{2(L + W)} = \frac{2(L \times W)}{(L + W)} \quad \dots(6.1.9)$$

(c) For hollow pipe having outer diameter, D and internal diameter, d

(Refer Fig. 6.1.2)

$$\begin{aligned} D_h &= 4 \times \frac{\pi}{4} (D^2 - d^2) \times \frac{1}{(\pi D + \pi d)} \\ &= \frac{(D + d)(D - d)}{(D + d)} \\ D_h &= (D - d) \quad \dots(6.1.10) \end{aligned}$$

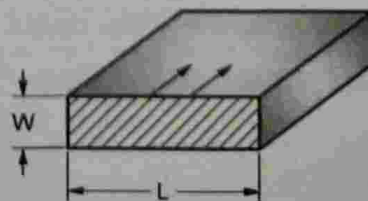


Fig. 6.1.1 : Flow through rectangular section

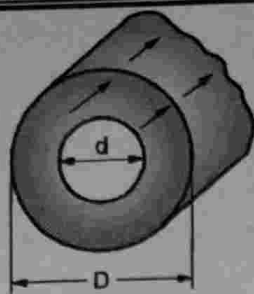


Fig. 6.1.2 : flow through annular space between two circular pipes

4. Bulk Mean temperature, T_m :

Properties of fluids are taken at bulk mean temperature given as :

$$T_m = \frac{T_{\text{inlet}} + T_{\text{outlet}}}{2} \quad \dots(6.1.11)$$

6.1.1 Co-relation for Flow of Liquid Metals in Pipes

When rate of heat transfer are required to be very high, liquid metals like potassium, sodium, mercury are used e.g. in Nuclear reactors.

Liquid metals have very low Prandtl number ($Pr < 0.01$). The co-relations are given as :

$$Nu = 5 + 0.025 (Re \cdot Pr)^{0.8} \quad \dots(6.1.12)$$

(For constant wall temperature)

$$Nu = 4.82 + 0.0185 (Pe)^{0.827} \quad \dots(6.1.13)$$

(For constant heat flux)

Liquid metals have very small thickness of velocity boundary layer compared to thermal boundary layer since Pr is small.

6.1.2 Co-relation for Flow of Heavy Oils in Pipes

According to Sieder and Tate relations :

$$Nu = 0.027 \cdot Re^{0.8} \cdot Pr^{0.33} (\mu / \mu_w)^{0.14} \quad \dots(6.1.14)$$

Where, μ_w is the dynamic viscosity of oil at wall temperature.

Note that the heavy oils have very high value of Prandtl number, Pr . Above relation is applicable both for constant wall temperature and constant heat flow conditions.

6.2 Co-relations for Fluid Flow Over the Plate

We have already discussed the following relationships in sections 5.6 and 5.7.

$$Re = \frac{\rho V D}{\mu} = \frac{V D}{\nu}$$

$$\text{Heat flux, } q = h_L (T_s - T_\infty)$$

$$\text{Total heat transfer rate, } Q = h \cdot A \cdot (T_s - T_\infty)$$

$$\text{where } h = \frac{1}{L} \int_0^L h_L \cdot dx$$

Other co-relations are as follows :

$$\frac{\text{Thermal boundary layer, } \delta_t}{\text{Velocity boundary layer, } \delta} = \frac{1}{1.026 \times Pr^{1/3}} \quad \dots(6.2.1)$$

1. Case I : Laminar flow ($Re < 3 \times 10^5$)

(a) Local Nusselt's number (at a distance, x from leading edge)

$$Nu_x = 0.332 (Re_x)^{1/2} \cdot (Pr)^{1/3} \quad \dots(6.2.2)$$

(b) Average heat transfer coefficient h_{av} and average Nusselt number $(Nu)_{av}$

$$Nu_x = \frac{h_x \cdot x}{k} = 0.332 (Re_x)^{1/2} \cdot (Pr)^{1/3} \quad \dots(i)$$

$$h_{av} = \frac{1}{L} \int_0^L h_x \cdot dx = \frac{1}{L} \int_0^L \left[0.332 k \left(\frac{V}{\nu} \right)^{1/2} \cdot Pr^{1/3} \cdot x^{-1/2} \right] dx$$

$$h_{av} = \frac{1}{L} \left[0.332 k \left(\frac{V}{\nu} \right)^{1/2} \cdot Pr^{1/3} \cdot \frac{x^{1/2}}{(1/2)} \right]_0^L$$

$$h_{av} = 0.332 \frac{k}{L} \left(\frac{V \cdot L}{\nu} \right)^{1/2} \cdot Pr^{1/3} \times 2 = 2 h_L \quad \dots(6.2.3)$$

$$\therefore Nu_{av} = \frac{h_{av} \cdot L}{k} = 0.664 \cdot Re^{1/2} \cdot Pr^{1/3} \quad \dots(6.2.4)$$

(c) Co-relation for liquid metals :

$$Nu = 1.13 (Re \cdot Pr)^{0.5} \quad \dots(6.2.5)$$

(for constant wall temperature)

2. Case II : For turbulent flow when ($Re > 5 \times 10^5$) and ($0.6 < Pr < 60$) :

$$Nu_x = 0.029 \cdot Re_x^{0.8} \cdot Pr^{1/3} \quad \dots(6.2.6)$$

$$(Nu)_{av} = 0.0366 Re^{0.8} \cdot Pr^{1/3} \quad \dots(6.2.7)$$

$$\text{Mean film temperature, } T_m = \frac{T_s + T_\infty}{2} \quad \dots(6.2.8)$$

Statement "For laminar/turbulent flow over a flat plate, the heat transfer coefficient decreases with the distance from the leading edge" is true.

For flow over flat plate,

$$Nu_x = 0.332 Re_x^{1/2} Pr^{1/3}$$

$$\text{or } h_L = 0.664 \frac{k}{L} (Re_L)^{1/2} Pr^{1/3}$$

$$= 0.664 \frac{k}{L} \left(\frac{VL}{\nu} \right)^{1/2} Pr^{1/3}$$

$$h_L = 0.664 \frac{k}{L^{1/2}} \cdot \left(\frac{V}{\nu} \right)^{1/2} Pr^{1/3}$$

In the above expression, as 'L', the length of plate along the flow increases, value of h_L decreases. ...Ans.

6.3 Co-relation for Flow Across a Horizontal Cylinder

(a) For constant heat flux :

$$Nu_D = 0.62 Re_D^{0.505} \dots(6.3.1)$$

(b) For constant wall temperature :

Hilpert's co-relation is :

$$Nu_D = C \times Re_D^n \cdot Pr^{1/3} \dots(6.3.2)$$

Value of 'C' and 'n' are given in Table 6.3.1.

Table 6.3.1 : Values of constants for Hilpert's correlations

Re_D	C	n
0.4 - 4	0.989	0.330
4 - 40	0.911	0.385
40 - 4000	0.615	0.466
4000 - 40000	0.174	0.618
40000 - 400000	0.0237	0.805

Table 6.3.2 : List of Formulae to be used for forced convection

Flow over flat plate	Flow through pipes / conduits
1. Laminar flow ($Re < 3 \times 10^5$)	1. Laminar flow ($Re < 2000$)
(i) Reynold's number., $Re = \frac{\rho \cdot V_{\infty} \cdot x}{\mu} = \frac{V_{\infty} \cdot x}{\nu}$	(i) $Re = \frac{\rho \cdot V_m \cdot D}{\mu} = \frac{V_m \cdot D}{\nu}$ (circular pipe) $Re = \frac{V_m \cdot D_h}{\nu}$ (Non-circular pipes) Hydraulic diameter $D_h = \frac{4 A_c}{P}$
(ii) Critical length $L_c = (3 \times 10^5) \cdot \frac{\nu}{V}$	(ii) Entry length, $L_c = 0.0575 Re \cdot D$
(iii) Average drag or friction coefficient, $C_f = \frac{1.328}{\sqrt{Re}} = 2 C_{fx}$	(iii) Average velocity, $V_m = \frac{V_{max}}{2}$
(iv) Drag force, $F_D = \frac{\rho \cdot A \cdot V^2}{2} \times C_f$	(iv) Friction factor, $f = \frac{16}{Re} = \frac{\Delta p}{4 \left(\frac{L}{D} \right) \left(\frac{\rho \cdot V^2}{2} \right)}$ where, $\Delta p = \frac{32 \mu V L}{D^2}$
(v) Velocity boundary layer thickness at distance 'x' from leading edge, $\delta_x = \frac{4.64 x}{\sqrt{Re}}$	
(vi) Mass flow rate, $m = \frac{5}{8} \rho \cdot V \cdot \delta_x$	



Flow over flat plate	Flow through pipes / conduits
2. Turbulent flow ($Re > 5 \times 10^5$) (i) $C_f = \frac{0.455}{\ln(Re_L^{2.58})} - \frac{C_1}{Re_L}$ where, $C_1 = 1050$ for change from laminar to turbulent flow $C_{fav} = 0.072 (Re_L)^{-0.2}$	2. Turbulent flow ($Re > 4000$) (i) Friction factor, $f = 0.046 (Re)^{-0.2} = \frac{\Delta p}{4 \left(\frac{L}{D}\right) \left(\frac{\rho \cdot V^2}{2}\right)}$
(ii) $\delta = \frac{0.39 x}{(Re)^{0.2}}$	(ii) Length of thermal entry region, $(Le)_t = 10 < \left(\frac{Le}{D}\right) \leq 60$ where hydrodynamic layer, $Le = 0.0575 Re_D \cdot D$
(iii) Ratio of thermal boundary layer thickness, δ_t to velocity boundary layer thickness, δ : $\frac{\delta_t}{\delta} = \frac{1}{1.026 Pr^{1/3}}$	
3. Dimensionless Numbers Reynold's No. : $Re_L = \frac{\rho \cdot V L}{\mu}$ Prandtl No. : $Pr = \frac{\mu \cdot C_p}{k_f}$ Nusselt No. : $Nu = \frac{h \cdot L}{k_f}$ Stanton No. : $St = \frac{(Nu)}{Re \cdot Pr} = \frac{h}{\rho V C_p}$ Peclet No. : $Pe = Re \cdot Pr$	3. Dimensionless Numbers Reynold's No. : $Re_L = \frac{\rho \cdot V D_R}{\mu}$ Prandtl No. : $Pr = \frac{\mu \cdot C_p}{k_f}$ Nusselt No. : $Nu = \frac{h \cdot D_R}{k_f}$ Stanton No. : $St = \frac{(Nu)}{Re \cdot Pr} = \frac{h}{\rho V C_p}$ Peclet No. : $Pe = Re \cdot Pr$
4. Heat transfer co-relation (A) Laminar flow : (i) Local Nusselt Number at distance x from leading edge. $Nu_x = 0.332 (Re_x)^{1/2} \cdot (Pr)^{1/3}$ (ii) Average heat transfer coefficient, $Nu_x = \frac{h_x \cdot x}{k} = 0.332 (Re_x)^{1/2} \cdot (Pr)^{1/3}$ $h_{av} = 2 \times h_L$ $h_{av} = \frac{Nu_{av} \cdot k}{L}$ $Nu_{av} = 0.664 (Re_x)^{1/2} \cdot (Pr)^{1/3}$ (iii) For liquid metal : $Nu = 1.13 (Re Pr)^{0.5}$	4. Heat transfer co-relation (A) Laminar flow : For constant heat flux : $Nu = 4.36$ For constant wall temperature : $Nu = 3.66$

Flow over flat plate	Flow through pipes / conduits
(B) Turbulent flow (i) Local Nusselt Number $Nu_x = 0.029 Re_x^{0.8} \cdot Pr^{1/3}$ $Nu_{av} = 0.0366 Re_x^{0.8} \cdot Pr^{1/3}$ (ii) Mean film temperature, $T_m = \frac{T_s + T_\infty}{2}$	(B) Turbulent flow $Nu = 0.023 Re^{0.8} \cdot Pr^{0.4} \text{ (When fluid is heated)}$ $Nu = 0.023 Re^{0.8} \cdot Pr^{0.3} \text{ (When fluid is cooled)}$
(C) Flow across a cylinder For const. Heat flux : $Nu_D = 0.62 Re_D^{0.505}$ For constant wall temperature : $Nu_D = C \times (Re_D)^n \cdot (Pr)^{1/3}$ For values of 'n' and 'C', Table 6.3 be referred	(C) Bulk mean temperature, $T_m = \frac{T_{inlet} + T_{outlet}}{2} \left(\text{or, } \frac{T_i + T_o}{2} \right)$
	(D) For flow of liquid metals ($Pr < 0.01$) $Nu = 5 + 0.025 (Re Pr)^{0.8} \text{ (for } T_w \text{ is constant)}$ $Nu = 4.82 + 0.0185 (Pe)^{0.827} \text{ (for constant heat flux)}$
	(E) For flow of heavy oils $Nu = 0.027 \cdot Re_x^{0.8} \cdot Pr^{0.33} (\mu / \mu_w)^{0.14}$ Where μ_w = dynamic viscosity at wall temperature.

Ex. 6.3.1 : What is meant by "hydraulic diameter" of a duct ? Explain its significance in convection heat transfer.

Calculate the hydraulic diameter for the following

Ducts :

- rectangular cross-sectioned, with sides = 10 cm and 5 cm
- square cross-sectioned, with side = 2 cm
- circular cross-sectioned with radius = 2cm.

Soln. : Theory : Refer Section 6.1(3)

Numerical

- (i) Hydraulic diameter, D_h for rectangular duct when $a = 10$ cm, $b = 5$ cm:

$$D_h = \frac{4 \times \text{Area of cross-section } A}{\text{Wetted perimeter } P} \quad \dots(i)$$

$$D_h = \frac{4(a \times b)}{2(a+b)} = \frac{4(10 \times 5)}{2(10+5)} = 6.667 \text{ cm} \dots \text{Ans.}$$

- (ii) D_h for square duct when $a = b = 2$ cm

$$D_h = \frac{4(a \times b)}{2(a+b)} = \frac{4 \times 2 \times 2}{2(2+2)} = 2 \text{ cm} \dots \text{Ans.}$$

- (iii) D_h for circular duct of radius $R = 2$ cm

From Equation (i),

$$D_h = \frac{4A}{P} = \frac{4 \times \pi R^2}{2\pi R} = 2R = 2 \times 2 = 4 \text{ cm}$$

(For circular duct, $D_h = D$)

...Ans.

Ex. 6.3.2 : A horizontal pipe 1 foot (0.3048 m) in diameter is maintained at a temperature of 250°C in a room where the ambient air is at 15°C . Calculate the free convection heat loss per meter of length.

Soln. :

$$D = 0.3048 \text{ m}, \quad T_s = 250^\circ \text{C}, \quad T_\infty = 25^\circ \text{C}$$

$$\Delta t = T_s - T_\infty = 250 - 25 = 225^\circ \text{C}; \quad L = 1 \text{ m}$$

Free convection heat loss per meter length, Q

Average heat transfer coefficient,

$$\bar{h} = 1.32 \left(\frac{\Delta t}{D} \right)^{1/4} = 1.32 \left(\frac{250 - 25}{0.3048} \right)^{1/4}$$

$$= 6.9556 \text{ W/m}^2\text{C}$$

$$Q = \bar{h} \times A \times (T_s - T_\infty) = \bar{h} \times \pi DL (T_s - T_\infty)$$

$$= 6.9556 \times (\pi \times 0.3048 \times 1) \times (250 - 25)$$

$$= 1565.2 \text{ W/m length} \dots \text{Ans.}$$



Ex. 6.3.3 : Air at 2 atm and 200° C is heated as it flows at a velocity of 12 m/s through a tube with a diameter of 3 cm. A constant heat flux condition is maintained at the wall and the wall temperature is 20° C above the air temperature all along the length of the tube. Calculate :

- The heat transfer per unit length of the tube.
- The increase in bulk temperature of air over a 4 m length of the tube. Properties of air $Pr = 0.681$, $\mu = 2.57 \times 10^{-5}$ kg/ms, $k = 0.0386$ W/mK,

$$C_p = 1.025 \text{ kJ/kgK}, \rho = 1.493 \text{ kg/m}^3, \text{ Use : } \overline{Nu} = 0.023 Re^{0.8} Pr^{0.4}$$

Soln. :

$$\begin{aligned} p &= 2 \text{ atm}, & T_s &= 200^\circ \text{ C}, & V &= 12 \text{ m/s}, \\ D &= 3 \text{ cm} = 0.03 \text{ m} \\ \Delta T &= 20^\circ \text{ C}, & Pr &= 0.681, & \mu &= 2.57 \times 10^{-5} \text{ kg/ms}, \\ k &= 0.0386 \text{ W/mK} \\ C_p &= 1.025 \text{ kJ/kgK}, & \rho &= 1.493 \text{ kg/m}^3, \\ \overline{Nu} &= 0.023 Re^{0.8} Pr^{0.4} \end{aligned}$$

Find : (a) Re using $\frac{\rho V D}{\mu}$

(b) $\overline{Nu}$

(c) h using $\overline{Nu} = \frac{h D}{k}$

- Heat transfer per unit length of tube, Q :

$$Re = \frac{\rho V D}{\mu} = \frac{1.493 \times 12 \times 0.03}{2.57 \times 10^{-5}} = 0.20914 \times 10^5$$

$$\overline{Nu} = \frac{h D}{k} = 0.023 Re^{0.8} Pr^{0.4}$$

$$\frac{h \times 0.03}{0.0386} = 0.023 \times (0.20914 \times 10^5)^{0.8} \times (0.681)^{0.4}$$

$$h = 72.577 \text{ W/m}^2\text{K}$$

$$\begin{aligned} Q &= h \cdot A_s \cdot \Delta T = h \times (\pi D L) \times \Delta T \\ &= 72.577 \times (\pi \times 0.03 \times 1) \times 20 \\ &= 136.8 \text{ W/m length of tube} \quad \dots \text{Ans.} \end{aligned}$$

- Increase in bulk temperature of tube over 4 m length of tube, $(\Delta T)_b$

Total heat transfer rate over 4 m tube, $Q_1 = 4Q$

$$Q_1 = \dot{m} C_p (\Delta T)_b = (\rho A V) \times C_p \times (\Delta T)_b$$

$$4Q = \rho A V \times C_p \times (\Delta T)_b$$

$$= \rho \times \frac{\pi}{4} D^2 \times V \times C_p \times (\Delta T)_b$$

$$4 \times 136.8 = 1.493 \times \frac{\pi}{4} \times (0.03)^2 \times 12 \times 1.025 \times (\Delta T)_b$$

$$(\Delta T)_b = 42.155^\circ \text{ C}$$

...Ans.

Ex. 6.3.4 : Air at 27° C and 1 atm flows over a flat plate at a velocity 3 m/s. The plate is heated over its entire length to a temperature of 70° C. Calculate the heat transferred if the plate length is 45 cm and width is 1 m. Properties of air, $v = 17.36 \times 10^{-6}$ m²/s, $k = 0.02749$ W/mK, $C_p = 1.006$ kJ/kg K, $Pr = 0.7$. Take $\overline{Nu}_L = 0.664 Re_L^{0.5} Pr^{1/3}$.

Soln. :

$$T_\infty = 27^\circ \text{ C}, \quad V = 3 \text{ m/s}, \quad T_s = 70^\circ \text{ C}$$

$$L = 45 \text{ cm} = 0.45 \text{ m} (x = L = 0.45 \text{ m}),$$

Width of plate, $W = 1 \text{ m}$

$$\nu = 17.36 \times 10^{-6} \text{ m}^2/\text{s}$$

$$k = 0.02749 \text{ W/mK}$$

$$C_p = 1.006 \text{ kJ/kg K}$$

$$Pr = 0.7$$

$$\text{Take } \overline{Nu}_L = 0.664 Re_L^{0.5} Pr^{1/3}$$

Area of plate,

$$A = L \times W = 0.45 \times 1 = 0.45 \text{ m}^2$$

Find : (a) Re to check whether flow is laminar or turbulent.

$$(b) \overline{Nu}_L, \text{ hence the value of } h_L \text{ using } \overline{Nu}_L = \frac{h \cdot L}{k}$$

$$(c) \text{ Heat transfer, } Q \text{ using } Q = h \cdot A (T_s - T_\infty)$$

$$Re_L = \frac{V \cdot L}{\nu} = \frac{3 \times 0.45}{17.36 \times 10^{-6}}$$

$$= 7.7765 \times 10^4 \text{ (flow is laminar)}$$

$$\overline{Nu}_L = 0.664 Re_L^{0.5} Pr^{1/3}$$

$$= 0.664 \times (7.7765 \times 10^4)^{0.5} \times (0.7)^{1/3} = 164.41$$

$$\overline{Nu}_L = \frac{h \cdot L}{k}, \quad h = \frac{\overline{Nu}_L \cdot k}{L}$$

$$h = \frac{164.41 \times 0.02749}{0.45} = 10.01 \text{ W/m}^2\text{K}$$

$$Q = h \times A \times (T_s - T_\infty) = h \times A \times (T_s - T_\infty)$$

$$= 10.01 \times 0.45 \times (70 - 27)$$

$$= 193.7 \text{ W}$$

...Ans.

Ex. 6.3.5 : Air at 2 atm and 200°C is heated as it flows at a velocity of 12 m/s through a tube with a diameter of 3 cm. A constant heat flux condition is maintained at the wall and the wall temperature is 20°C above the air temperature all along the length of the tube. Calculate

- (a) the heat transfer per unit length of the tube.
 (b) the increase in bulk temperature of air over a 4m length of the tube.

Properties of air at 200°C are $Pr = 0.681$,
 $\mu = 2.57 \times 10^{-5} \text{ kg/ms}$,
 $k = 0.0386 \text{ W/mK}$ and $C_p = 1.025 \text{ kJ/kg} \cdot \text{K}$.
 Use : $Nu_d = 0.023 (Re_d)^{0.8} (Pr)^{0.4}$

Soln. :

Given : Temperature of air, $T_\infty = 200^\circ\text{C}$; velocity of air,
 $V = 12 \text{ m/s}$

Tube diameter, $D = 3 \text{ cm} = 0.03 \text{ m}$;

$p = 2 \text{ atm} = 2 \times 101.325 \text{ kPa}$

Wall temperature, $T_s = 20^\circ\text{C} + T_\infty = 20 + 200 = 220^\circ\text{C}$

Find :

- (a) Density of air using gas equation since it is not given
 (b) Re (c) Nu (d) h (e) Q/L
 (f) Bulk temperature rise using the concept;

Rate of change in internal energy = Rate of heat transfer.

$$\begin{aligned} \text{Density of air, } \rho &= \frac{p}{RT} \\ &= \frac{101.325 \times 2}{0.287 \times 473} = 1.493 \text{ kg/m}^3 \\ Re_D &= \frac{\rho V D}{\mu} = \frac{1.493 \times 12 \times 0.03}{2.57 \times 10^{-5}} \\ &= 20914 > 4000 \end{aligned}$$

$$\begin{aligned} \text{Hence, flow is turbulent } Nu &= 0.023 (Re_D)^{0.8} (Pr)^{0.4} \\ &= 0.023 (20914)^{0.8} (0.681)^{0.4} = 56.404 \end{aligned}$$

$$\text{But, } Nu = \frac{h \cdot D}{k};$$

$$h = \frac{Nu \cdot k}{D} = \frac{56.404 \times 0.0386}{0.03} = 72.57 \text{ W/m}^2\text{K}$$

(a) Heat transfer rate per unit length, (Q/L) :

$$Q = h \cdot (\pi D L) (T_s - T_\infty)$$

$$\begin{aligned} \frac{Q}{L} &= h \pi D \times (T_s - T_\infty) = 72.57 \times \pi \times 0.03 (20) \\ &= 136.79 \text{ W/m} \end{aligned}$$

...Ans.

(b) Increase in bulk temperature over a 4m length tube :

$$\dot{m} = \rho \cdot A \cdot V = \rho \frac{\pi}{4} D^2 \cdot V$$

$$= 1.493 \times \frac{\pi}{4} \times (0.03)^2 \times 12 = 0.012664 \text{ kg/s}$$

Rate of change in internal energy = Rate of heat transfer
 /unit length $\left(\frac{Q}{L}\right) \times L$

$$\dot{m} C_p \cdot \Delta T_b = \frac{\dot{Q}}{L} \times L$$

$$0.012664 \times (1.025 \times 10^3) \Delta T_b = 136.79 \times 4$$

$$\Delta T_b = 42.15^\circ\text{C}$$

...Ans.

Ex. 6.3.6 : Calculate the coefficient of heat transfer by free convection and the maximum current intensity for a nichrome wire 2.5 mm in diameter with the condition that its temperature should not exceed 95°C. The wire is exposed to still air at 25°C and the resistance per meter length of the wire is 6 ohm/m. Use $Nu = 1.18 (GrPr)^{1/8}$. The properties of air at 60°C are $\nu = 18.97 \times 10^{-6} \text{ m}^2/\text{s}$, $Pr = 0.696$, $k = 0.02896 \text{ W/mK}$.

Soln. : Refer Fig P.6.3.6

Given :

$$D = 2.5 \text{ mm} = 0.0025 \text{ m};$$

$$T_s = 95^\circ\text{C}; \quad T_\infty = 25^\circ\text{C}$$

$$R = 6 \Omega/\text{m}$$

$$Q = I^2 R \quad \dots(i)$$

$$\text{and } Q = hA (T_s - T_\infty) \quad \dots(ii)$$

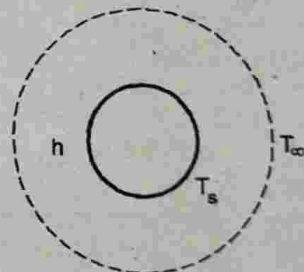


Fig. P. 6.3.6

$$\text{Mean temperature, } T_m = \frac{95 + 25}{2} = 60 + 273 = 333 \text{ K}$$

$$\begin{aligned} Gr &= \frac{g \beta \Delta T D^3}{\nu^2} = \frac{9.81 \times 1 \times (95 - 25) \times 0.0025^3}{333 \times (18.97 \times 10^{-6})^2} \\ &= 89.54 \end{aligned}$$

Now h can be found out from given relation

$$Nu = 1.18 (Gr \cdot Pr)^{1/8} = 1.18 (Gr \cdot Pr)^{0.125}$$

$$\text{and } Nu = \frac{h \cdot D}{k}$$

$$Nu = \frac{hD}{k} = 1.18 (Gr \cdot Pr)^{0.125}$$

$$h = \frac{0.02896 \times 1.18}{0.0025} (89.54 \times 0.696)^{0.125}$$



$$= 22.9 \text{ W/m}^2\text{K} \quad \dots \text{Ans.}$$

$$\text{Now, } Q = hA(T_s - T_\infty)$$

$$= 22.9 \times \pi \times 0.0025 \times 1 \times (95 - 25) = 12.58 \text{ W}$$

$$\therefore Q = I^2 R$$

$$I = \sqrt{\frac{Q}{R}} = \sqrt{\frac{12.58}{6}} = 1.45 \text{ Amp.} \quad \dots \text{Ans.}$$

Ex. 6.3.7 : An electrical air heater consists of a horizontal array of thin metal strips that are each 10 mm long in the direction of an air stream that is in parallel flow over the top of the strips. Each strip is 0.2 m wide, and 25 strips are arranged side by side, forming a continuous and smooth surface over which the air flows at 2 m/s. Each strip is maintained at 500°C and the air is at 25°C.

What is the rate of convection heat transfer from the first strip? The fifth strip? All the strips?

For air at 535 K, take : $\nu = 43.54 \times 10^{-6} \text{ m}^2/\text{s}$; $k = 0.0429 \text{ W/mK}$; $Pr = 0.683$.

Soln. : Air properties at mean temperature

$$T_m = \frac{500 + 273 + 25 + 273}{2} = 535.5 \text{ K}$$

$$\nu = 43.54 \times 10^{-6} \text{ m}^2/\text{s}; k = 0.0429 \text{ W/mK.}$$

$$Pr = 0.683$$

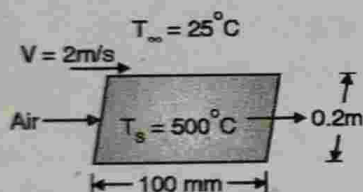


Fig. P. 6.3.7



First strip :

Fig. P. 6.3.7(a)

$$Re_L = \frac{VL}{\nu} = \frac{2 \times 0.01}{43.54 \times 10^{-6}} = 459.3$$

Since $Re_L = 459.3$ and $Pr > 0.6$,

Flow is laminar, hence

$$Nu_x = 0.332 Re_x^{1/2} Pr^{1/3} \text{ shall be used}$$

$$\frac{hx}{k} = 0.332 \left(\frac{V \cdot x}{\nu} \right)^{1/2} Pr^{1/3}$$

$$\begin{aligned} h_{av} &= 0.332 \frac{k}{L} \left(\frac{V}{\nu} \right)^{1/2} \int_0^L x^{-1/2} Pr^{1/3} dx \\ &= 0.332 \frac{k}{L} \left(\frac{V}{\nu} \right)^{1/2} \left[\frac{x^{1/2}}{1/2} \right]_0^L \cdot Pr^{1/3} \end{aligned}$$

$$= 0.664 \frac{k}{L} \left(\frac{VL}{\nu} \right)^{1/2} Pr^{1/3}$$

$$\begin{aligned} h_{av} &= 0.664 \times \frac{0.0429}{0.01} \left(\frac{2 \times 0.01}{43.54 \times 10^{-6}} \right)^{1/2} (0.683)^{1/3} \\ &= 53.77 \text{ W/m}^2\text{K.} \end{aligned}$$

Rate of heat convection from first strip :

$$\begin{aligned} Q &= h A \Delta T \\ &= 53.77 \times (0.2 \times 0.01 \times 2) \times (500 - 25) \\ &= 102.16 \text{ W from both sides} \quad \dots \text{Ans.} \end{aligned}$$

From fifth strip :

$$L = 0.01 \times 5 = 0.05 \text{ m}$$

$$Re = \frac{VL}{\nu} = \frac{2 \times 0.05}{43.54 \times 10^{-6}} = 2296.74$$

$$\begin{aligned} h &= 0.664 \frac{k}{L} (Re)^{1/2} Pr^{1/3} \\ &= 0.664 \times \frac{0.0429}{0.05} (2296.74)^{1/2} \times (0.683)^{1/3} \\ &= 24 \text{ W/m}^2\text{K} \end{aligned}$$

$$\begin{aligned} Q &= h A \Delta T = 24 \times (0.2 \times 0.01 \times 2) \times (500 - 25) \\ &= 45.6 \text{ W} \quad \dots \text{Ans.} \end{aligned}$$

For all strips :

$$L = 0.01 \times 25 = 0.25 \text{ m}$$

$$\begin{aligned} (Re) &= \frac{2 \times 0.25}{43.54 \times 10^{-6}} \\ &= 11483.6 \text{ hence flow remains laminar} \end{aligned}$$

$$\begin{aligned} h &= 0.664 \frac{k}{L} Re^{1/2} Pr^{1/3} \\ &= 0.664 \times \frac{0.0429}{0.25} \left(\frac{2 \times 0.25}{43.54 \times 10^{-6}} \right)^{1/2} (0.683)^{1/3} \\ &= 10.75 \text{ W/m}^2\text{K} \end{aligned}$$

$$\begin{aligned} Q &= h A \Delta T \\ &= 10.75 \times (0.2 \times 0.25 \times 2) \times (500 - 25) \\ &= 510.6 \text{ W} \quad \dots \text{Ans.} \end{aligned}$$

Ex. 6.3.8 : Compare the relative merits of water under pressure and liquid sodium as coolant; when flowing through a stainless steel tube of 0.75 cm ID with a velocity of 5 m/s. The tube wall temperature is 300°C and temperature at the inlet to the pipe is 200°C.

Use the correlation : $Nu_D = 0.23 Re_D^{0.8} Pr^{0.33}$

Properties of water : $\nu = 0.16 \times 10^{-6} \text{ m}^2/\text{s}$

$Pr = 0.938$, $k = 0.665 \text{ W/mK}$,

Properties of liquid sodium : $\nu = 4.95 \times 10^{-7} \text{ m}^2/\text{s}$, $Pr = 0.0075$, $k = 80 \text{ W/mK}$.

Soln. :

Given : $V = 5 \text{ m/s}$, $D = 0.75 \text{ cm} = 0.75 \times 10^{-2} \text{ m}$

Aim is to find the value of h both for water and liquid sodium so as get their relative merits.

$$Re_D = \frac{VD}{\nu} = \left[\frac{5 (0.75 \times 10^{-2})}{0.16 \times 10^{-6}} \right] = 234375$$

For water : $Nu = 0.023 Re^{0.8} \cdot Pr^{0.33} = \frac{h \cdot D}{k}$

$$Nu = 0.023 \times (234375)^{0.8} \times (0.938)^{0.33} = 445.1$$

$$h = \frac{Nu \cdot k}{D} = \frac{445.1 \times 0.665}{0.75 \times 10^{-2}} = 39468 \text{ W/m}^2\text{K}$$

For liquid sodium : $Re = \frac{V \cdot D}{\nu} = \frac{5 (0.75 \times 10^{-2})}{(4.95 \times 10^{-7})} = 75757.6$

$$Nu = 0.023 Re^{0.8} \cdot Pr^{0.33} = 0.023 (75757.6)^{0.8} \times (0.0075)^{0.33} = 36.77$$

$$Nu = \frac{h \cdot D}{k}; h = \frac{Nu \cdot k}{D}$$

$$h = \frac{36.77 \times 80}{(0.75 \times 10^{-2})} = 3,92,213.3 \text{ W/m}^2\text{K} \dots \text{Ans.}$$

The heat transfer coefficient for liquid sodium is more as compared to water, thus the use of liquid sodium is beneficial where high rates of heat transfer are needed.

Ex. 6.3.9 : It is required to estimate the heat transfer from a cylinder 50 mm diameter and surface temperature 140°C when kept in a cross-flow of air at a velocity of 4 m/s and temperature 20°C . For this purpose, scale-model experiment are performed a 1/5 scale-model and following results are obtained from the experiments on the model :

Velocity of air (m/s)	2.0	5.0	10.0	20.0
Heat transfer coefficient (W/m ² K)	39.5	71.2	106.5	165.3

Calculate the heat transfer coefficient and the rate of heat transfer per meter length of the actual cylinder.

Soln. :

Given : Prototype : $D_p = 50 \text{ mm}$; $T_s = 140^\circ\text{C}$, $V = 4 \text{ m/s}$, $T_\infty = 20^\circ\text{C}$

Scale : $\frac{D_p}{D_m} = 5$ i.e. $D_m = \frac{D_p}{5} = \frac{50}{5} = 10 \text{ mm}$

For forced convection model testing, Reynold's number of the prototype and model must be equal.

$$\left(\frac{VD}{\nu} \right)_m = \left(\frac{VD}{\nu} \right)_p$$

Because the model testing is conducted at same surface and air temperature, ν can be considered to be constant :

$$\therefore (VD)_m = (VD)_p$$

$$V_m = \frac{(VD)_p}{D_m} = \frac{4 \times 50}{10} = 20 \text{ m/s}$$

The heat transfer coefficient at 20 m/s on the model is $165.3 \text{ W/m}^2\text{K}$

The heat transfer for actual cylinder per m length is ;

$$Q = h \cdot A (T_s - T_\infty) = 165.3 \times (\pi \times 0.05 \times 1) (140 - 20)$$

$$= 3116 \text{ W/m length} \dots \text{Ans.}$$

Ex. 6.3.10 : Heavy oil is pumped and flows over a heated plate ($0.4 \text{ m} \times 0.2 \text{ m}$ size) at 30°C and a velocity of 1.3 m/s . The flow of oil is parallel to the 0.4 m side of the plate. The plate is kept at a uniform temperature of 60°C . Calculate the rate of heat transfer from the plate to the oil.

The properties of oil at mean temperature of $\left(\frac{60 + 30}{2} = 45^\circ\text{C} \right)$ are :

$$\rho = 877 \text{ kg/m}^3, \quad \nu = 25 \times 10^{-5} \text{ m}^2/\text{s}, \quad k = 0.145 \text{ W/mK},$$

$$\mu = 0.2193 \text{ Ns/m}^2, \quad C_p \text{ of oil} = 1.97 \text{ kJ/kg K},$$

Assume $Nu = 4.36$ for Laminar flow.

Soln. :

Given : $L = 0.4 \text{ m}$, $W = 0.2 \text{ m}$, $T_\infty = 30^\circ\text{C}$, $V = 1.3 \text{ m/s}$, $T_s = 60^\circ\text{C}$.

Find :

(a) Re to check whether flow is laminar or turbulent.

(b) Nu (c) h (d) Q

$$Re = \frac{VL}{\nu} = \frac{1.3 \times 0.4}{25 \times 10^{-5}} = 0.0208 \times 10^5$$

$\therefore Re < 3 \times 10^5$ flow is laminar.

$$\therefore Nu = 4.36 = \frac{h \cdot L}{k}, \quad 4.36 = \frac{h \times 0.4}{0.145}$$

$$h = 1,5805 \text{ W/m}^2\text{K}$$

Rate of heat transfer, Q :

$$Q = h \cdot A \cdot (T_s - T_\infty) = 1,5805 \times (0.4 \times 0.2) (60 - 30) = 3.79 \text{ W}$$

$\dots \text{Ans.}$

Ex. 6.3.11 : The engine of an automobile can be considered as a box of length of 0.7 m , width 0.23 m and height 0.1 m . Its surface temperature is maintained at 60°C . When the



automobile moves at a velocity of 30 m/s, the air at 20°C passes over the crankcase at relative speed. Assuming the flow to be turbulent with $Nu_L = 0.036 Re_L^{0.8} Pr^{1/3}$, find the rate of heat transfer from all sides of the crankcase.

Take : $\rho = 1.09 \text{ kg/m}^3$, $\nu = 20.1 \times 10^{-6}$, $k = 0.027 \text{ W/mK}$, $Pr = 0.7$

Soln. :

Given : $L = 0.7 \text{ m}$, $W = 0.23 \text{ m}$, $H = 0.1 \text{ m}$
 $T_s = 60^\circ\text{C}$, $V = 30 \text{ m/s}$, $T_\infty = 20^\circ\text{C}$,

Find : (a) Re_L (b) Nu_L (c) h (d) Q .

$$Re_L = \frac{\rho \cdot V \cdot L}{\mu} = \frac{1.09 \times 30 \times 0.7}{20.1 \times 10^{-6}} = 1.1388 \times 10^6$$

$$Nu_L = 0.036 (Re_L)^{0.8} (Pr)^{1/3}$$

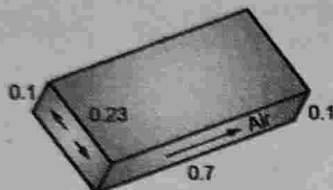


Fig. P. 6.3.11

$$= 0.036 (1.1388 \times 10^6)^{0.8} \times (0.7)^{1/3} = 2237.8$$

$$\text{But, } Nu_L = \frac{h \cdot L}{k};$$

$$h = \frac{Nu_L \cdot k}{L} = \frac{2237.8 \times 0.027}{0.7} = 86.32 \text{ W/m}^2\text{K}$$

Total surface area,

$$A = 2(L \times W) + 2(L \times H) + 2(W \times H)$$

$$= 2(0.7 \times 0.23) + 2(0.7 \times 0.1) + 2(0.23 \times 0.1) = 0.508 \text{ m}^2$$

$$Q = h \cdot A (T_s - T_\infty)$$

$$= 86.32 \times 0.508 \times (60 - 20) = 1754 \text{ W} \dots \text{Ans.}$$

Ex. 6.3.12 : A flat plate is 2m long, 0.8m wide and 3 mm thick. Density of plate = 3,000 kg/m³. Specific heat of plate material = 700 J/kg K. Its initial temperature is 80°C. A stream of air at 20°C is blown over both surfaces of the plate along its width, at a velocity 2 m/s. Calculate,

- rate of heat dissipation from the plate,
- initial rate of cooling of the plate,

Properties of air :

$$\rho = 1.09 \text{ kg/m}^3, \quad k = 0.028 \text{ W/mK}, \quad C_p = 1.007 \text{ kJ/kgK}$$

$$\mu = 2.03 \times 10^{-5} \text{ kg/ms}, \quad Pr = 0.698 \quad Nu = 0.664 \quad Re^{0.5}$$

$$Pr^{0.3333}$$

Soln. : Mean temperature of air,

$$T_m = \frac{80 + 20}{2} = 50^\circ\text{C}$$

Given : $L = 2 \text{ m}$, $W = 0.8 \text{ m}$,
 $t = 3 \text{ mm} = 3 \times 10^{-3} \text{ m}$.

Density of plate, $\rho_1 = 3000 \text{ kg/m}^3$, $C_{p1} = 700 \text{ J/kg K}$,
 $T_s = 80^\circ\text{C}$, $T_\infty = 20^\circ\text{C}$,

Find :

- Re , check type of flow
- Nu
- h
- Q
- Rate of cooling

(i) Rate of heat dissipation from the plate

$$\text{Reynold's number, } Re = \frac{\rho VL}{\mu} = \frac{1.09 \times 2 \times 0.8}{2.03 \times 10^{-5}}$$

$$= 8.59 \times 10^4$$

($L = 0.8 \text{ m}$ to be taken along the flow)

$\therefore Re < 3 \times 10^5$ the flow is laminar.

$$Nu = 0.664 (Re)^{0.5} (Pr)^{0.3}$$

$$= 0.664 \times (8.59 \times 10^4)^{0.5} (0.698)^{0.3} = 174.7$$

$$Nu = \frac{h \cdot L}{k}; \quad h = \frac{Nu \times k}{L}$$

$$= \frac{174.7 \times 28.26 \times 10^{-3}}{0.8} = 6.11 \text{ W/m}^2\text{K}$$

(i) Rate of heat dissipation, Q :

$$Q_1 = h A (T_s - T_\infty)$$

$$= 6.11 \times 2 \times 0.8 \times (80 - 20) = 586.56 \text{ W}$$

As Air is blowing on both side of a plate

$$Q_1 = 2 \times 586.56 = 1173 \text{ W} \dots \text{Ans.}$$

(ii) Initial rate of cooling :

The heat lost by the plate instantaneously is given by,

$$Q = m C_p \Delta T \dots (i)$$

$$\text{Mass of plate, } m = (A \times \text{thickness}) \times \rho_1$$

$$= 2 \times 0.8 \times 0.003 \times 3000$$

$$= 14.4 \text{ kg}$$

$$\text{From Equation (i)} \quad 1173 = 144 \times 700 \times (\Delta T)$$

$$\therefore \Delta T = 0.116^\circ\text{C/S} \dots \text{Ans.}$$

Ex. 6.3.13 : Liquid metal at the rate of 5 kg/s flows in a tube of 6 cm diameter. It enters at 400°C into the tube and leaves at 430°C. A constant heat flux is maintained along the tube. The tube surface temperature is 22°C higher than the temperature of liquid metal. Calculate the length of tube required. Assume the following properties for the liquid metal :

$$\mu = 1.35 \times 10^{-3} \text{ kg/ms}, \quad C_p = 150 \text{ J/kg K}, \quad Pr = 0.011,$$

$$k = 16 \text{ W/mK}.$$

Correlation for constant heat flux is : $Nu = 4.82 + 0.0185 (Re \cdot Pr)^{0.83}$.

Soln. :

Given : $\dot{m} = 5 \text{ kg/s}$, $D = 6 \text{ cm} = 0.06 \text{ m}$,

$$T_s - T_\infty = 22^\circ\text{C}, T_i = 400^\circ\text{C}, T_o = 430^\circ\text{C}$$

Find : (a) Nu (b) h using $Nu = \frac{h \cdot D}{k}$

(c) V from continuity Equation

(d) Q_c

(e) use Q_c = change in internal energy to find L.

$$\dot{m} = \rho \cdot A \cdot V;$$

$$\dot{m} = \rho \times \frac{\pi}{4} D^2 \times V$$

$$\therefore 5 = \rho \times \frac{\pi}{4} \times (0.06)^2 \times V; V = \frac{1768.4}{\rho} \quad \dots(i)$$

$$Re = \frac{\rho \cdot V D}{\mu}$$

$$= \rho \times \frac{1768.4}{\rho} \times 0.06 \times \frac{1}{1.35 \times 10^{-3}} = 78595$$

$$\therefore Nu = 4.82 + 0.0185 (Re \cdot Pr)^{0.83}$$

$$= 4.82 + 0.0185 (78595 \times 0.011)^{0.83} = 9.886$$

But, $Nu = \frac{h \cdot D}{k}; 9.886 = \frac{h \times 0.06}{16}$

$$h = 2636.27 \text{ W/m}^2 \text{ K}$$

Heat transfer rate by convection,

$$Q_c = h \cdot A \cdot (T_s - T_\infty) = h \cdot \pi D L (T_s - T_\infty)$$

$$= 2636.27 \pi \times 0.06 L \times 22$$

$$= 10932.35 L \text{ (Watts)}$$

This heat transfer by convection must be equal to change in internal energy. Therefore,

$$Q_c = \dot{m} \cdot C_p \cdot (T_o - T_i);$$

$$10932.35 L = 5 \times 150 \times (430 - 400)$$

$$L = 2 \text{ m} \quad \dots\text{Ans.}$$

Ex. 6.3.14 : 65 kg/min of water is heated from 30°C to 60°C by passing through duct of $3 \text{ cm} \times 2 \text{ cm}$. The duct is heated by condensing the steam on its outer surface. Find out the length of the duct required.

Properties of water : $\rho = 995 \text{ kg/m}^3$,

$\mu = 7.65 \times 10^{-4} \text{ kg/ms}$,

$C_p = 4.174 \text{ KJ/kg K}$, $k = 0.623 \text{ W/mK}$,

Conductivity of duct material = 35 W/mK .

Use the following correlations :

$$Nu = 0.023 Re^{0.8} Pr^{0.4} \text{ for turbulent flow}$$

$$Nu = 4.36 \text{ for laminar flow}$$

Soln. :

Given : $\dot{m} = 65 \text{ kg/min} = \frac{65}{60} \text{ kg/s} = 1.083 \text{ kg/s}$,

$$a = 3 \text{ cm}, b = 2 \text{ cm};$$

$$A = a \times b = 3 \times 2 = 6 \text{ cm}^2 = 6 \times 10^{-4} \text{ m}^2$$

Find : (a) D_h (b) V using $\dot{m} = \rho \cdot A \cdot V$

(c) Re_D (d) Pr (e) Nu (f) h

(g) L by using concept of rate of heat transfer = rate of change of internal energy.

The hydraulic diameter,

$$D_h = \frac{4A}{P} = \frac{4ab}{2(a+b)} = \frac{2 \times 3 \times 2}{(3+2)} = 2.4 \text{ cm}$$

$$\dot{m} = \rho A V$$

$$\therefore 1.083 = 995 \times (6 \times 10^{-4}) \times V; V = 1.8141 \text{ m/s}$$

$$Re_D = \frac{\rho \cdot V \cdot D_h}{\mu} = \frac{995 \times 1.8141 \times 2.4 \times 10^{-2}}{7.65 \times 10^{-4}}$$

$$= 5.6 \times 10^4$$

$$\therefore Re_D > 4000, \text{ flow is turbulent.}$$

$$Pr = \frac{\mu C_p}{k} = \frac{7.65 \times 10^{-4} \times 4.174 \times 10^3}{0.623} = 5.125$$

$$Nu = 0.023 Re^{0.8} Pr^{0.4}$$

$$= 0.023 (5.6 \times 10^4)^{0.8} \times (5.125)^{0.4} = 278.07$$

$$Nu = \frac{h \cdot D_h}{k}$$

$$278.07 = \frac{h \times (2.4 \times 10^{-2})}{0.623}$$

$$h = 7218.3 \text{ W/m}^2 \text{ K}$$

Heat transfer

$$Q = \dot{m} C_p \Delta T = 1.083 \times 4174 \times (60 - 30)$$

Heat transfer to water is taking place from condensing steam so heating temperature of duct surface is 100°C .

$$\text{Therefore } Q = h A \Delta T = 7218.3 \times 2 (0.03 + 0.02) \times L$$

$$\times \left(100 - \frac{30 + 60}{2} \right)$$

$$\text{Hence } Q = 1.083 \times 4174 \times (60 - 30) = 7218.3 \times 2$$

$$(0.03 + 0.02) \times L \times \left(100 - \frac{30 + 60}{2} \right)$$

$$\therefore L = 3.42 \text{ m} \quad \dots\text{Ans.}$$

Ex. 6.3.15 : Water flows at the rate of 0.1 kg/s through a thin metallic tube of 1 cm diameter and 3 m length. It enters the tube at 25°C . The outer surface of the tube maintained at a constant temperature of 100°C . Calculate the exit temperature of water.

$$Nu = 0.023 Re^{0.8} Pr^{0.4} \text{ for turbulent flow}$$

$$= 4.36 \text{ for laminar flow}$$



Properties of water : $C_p = 4174 \text{ J/kg K}$,

$$\mu = 5.62 \times 10^{-4} \text{ kg/ms}, \quad k = 0.644 \text{ W/mK}$$

Soln. :

Given : $\dot{m} = 0.1 \text{ kg/s}$, Tube diameter, $D = 1 \text{ cm}$
 $= 1 \times 10^{-2} \text{ m}$ Length of tube, $L = 3 \text{ m}$,
 Temperature of water at entry, $T_i = 25^\circ\text{C}$,
 $T_s = 100^\circ\text{C}$,
 Let exit temperature of water be T_o .

Find : (a) Velocity of water using $\dot{m} = \rho A V$

(b) Re (c) Pr (d) Nu (e) L.

Area of tube,

$$A = \frac{\pi}{4} (D^2) = \frac{\pi}{4} (1 \times 10^{-2})^2 = \frac{\pi}{4} \times 10^{-4} \text{ m}^2$$

$$\dot{m} = \rho \cdot A \cdot V;$$

$$0.1 = 1000 \times \left(\frac{\pi}{4} \times 10^{-4} \right) \cdot V$$

$$V = 1.2732 \text{ m/s}$$

Kinematic viscosity,

$$\text{Re} = \frac{V \cdot D}{\nu} = \frac{1.2732 \times (1 \times 10^{-2})}{5.62 \times 10^{-7}}$$

$$= 22655 \left(\because \nu = \frac{\mu}{\rho} \right)$$

Hence flow is turbulent as $\text{Re} > 4000$

$$\text{Pr} = \frac{\mu \cdot C_p}{k} = \frac{5.62 \times 10^{-4} \times 4174}{0.644} = 3.643$$

$$\text{Nu} = 0.023 \text{Re}^{0.8} \text{Pr}^{0.4}$$

$$= 0.023 (22655)^{0.8} (3.643)^{0.4} = 117.6$$

But, $\text{Nu} = \frac{h \cdot D}{k};$

$$117.6 = \frac{h \times (1 \times 10^{-2})}{0.644}$$

$$h = 7574 \text{ W/m}^2 \text{ K}$$

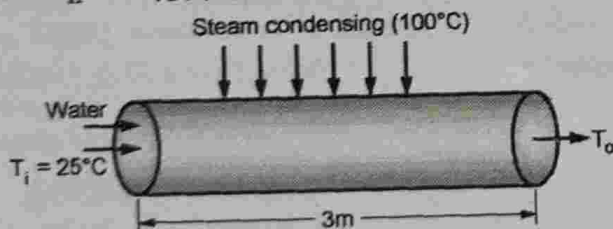


Fig. P. 6.3.15

Heat transfer

$$Q = \dot{m} C_p (T_o - T_i) = h A \Delta T$$

$$= h \cdot \pi D \cdot L \left(100 - \frac{T_o + 25}{2} \right)$$

Or $Q = 0.1 \times 4174 (T_o - 25)$

$$= 7574 \times \pi \times 0.01 \times 3 \left(100 - \frac{T_o + 25}{2} \right)$$

Or $417.4 T_o - 10435 = 62461 - 356.92 T_o$

Or $774.37 T_o = 72896$

$$T_o = 94^\circ\text{C}$$

...Ans.

Ex. 6.3.16 : A flat plate 2m long and 1m wide is maintained at constant temperature of 100°C . The air at 20°C blows over the plate along the length of plate. If the rate of heat transfer from plate is 4000 W, find the required velocity of air.

The air properties at the mean temperature of 60°C are :

$$\rho = 1.06 \text{ kg/m}^3, \quad \nu = 18.97 \times 10^{-6} \text{ m}^2/\text{s}, \quad \text{Pr} = 0.696,$$

$$k = 0.0283 \text{ W/mK}$$

Assume heat transfer from both side of the plate.

Take : $\text{Nu} = 0.664 \text{Re}^{1/2} \cdot \text{Pr}^{1/3}$ for laminar flow.

$$\text{Nu} = 0.0366 \text{Re}^{0.8} - 836 \text{Pr}^{1/3} \text{ for turbulent flow.}$$

Soln. :

Given : $L = 2 \text{ m}$, $W = 1 \text{ m}$, $T_s = 100^\circ\text{C}$,

$T_\infty = 20^\circ\text{C}$, $Q = 4000 \text{ W}$.

Find :

(a) Since blow is over plate, velocity can only be calculated from Re. Re can be found out from Nu relation provide type of flow is known. So we will have to assume and check.

(b) h from given value of Q .

(c) Using $\text{Nu} = \frac{h \cdot x}{k}$ find Nu. Using Nu find Re by assuming the flow to be either as laminar or turbulent and check its correctness.

(d) Find V using correct value of Re.

$$A = 2 (L \times W) = 2 (2 \times 1) = 4 \text{ m}^2$$

...($\because$ Heat is transferred from both sides of plate)

$$Q = h \cdot A \cdot (T_s - T_\infty)$$

$$4000 = h \times 4 \times (100 - 20);$$

$$h = 12.5 \text{ W/m}^2 \text{ K}.$$

$$\text{Nu} = 0.664 \text{Re}^{1/2} \cdot \text{Pr}^{1/3} = \frac{h \cdot x}{k} \dots (\text{where, } x = L)$$

; assuming laminar flow

$$\text{i.e. } 0.664 (\text{Re})^{1/2} (0.696)^{1/3} = \frac{12.5 \times 2}{0.0283}$$

$$\text{Re} = 2.2537 \times 10^6$$

(a) Since $\text{Re} > 5 \times 10^5$ it shows that flow is turbulent and assumption, made for laminar flow is wrong.

Hence, using the co-relation for turbulent flow,

$$Nu = 0.0366 (Re)^{0.8} - (Pr)^{1/3} \times 836 = \frac{h \cdot x}{k}$$

$$\text{i.e. } 0.0366 (Re)^{0.8} - (0.696)^{1/3} \times 836 = \frac{12.5 \times 2}{0.0283}$$

$$Re = 6.44124 \times 10^5$$

Since $Re > 5 \times 10^5$, assumption that flow is turbulent is correct.

$$Re = \frac{V \cdot L}{\nu} ; 6.44124 \times 10^5 = \frac{V \times 2}{18.97 \times 10^{-6}}$$

$$= 6.11 \text{ m/s}$$

...Ans.

Ex. 6.3.17 : Air flows in a copper tube of 2 cm diameter at a velocity of 30 m/s. The temperature of air at inlet and outlet of tube are 15°C and 270°C respectively.

The properties of air at mean temperature are :

$$\nu = 3.591 \times 10^{-5} \text{ m}^2/\text{s}, \quad Pr = 0.681, \quad k = 0.0394 \text{ W/mK},$$

$$C_p = 1027 \text{ J/kg K}, \quad \rho = 0.73 \text{ kg/m}^3.$$

The tube is maintained at a constant temperature of 280°C. Using the Reynolds analogy, calculate the length of tube and pumping power required.

Take friction factor, $f = 0.3165 (Re)^{-0.25}$ (for turbulent flow)

Soln. :

Given : $D = 2 \text{ cm} = 0.02 \text{ m}, \quad V = 30 \text{ m/s},$

$$T_i = 15^\circ\text{C} = 288 \text{ K},$$

$$T_o = 270^\circ\text{C} = 543 \text{ K}, \quad T_s = 280^\circ\text{C}.$$

Find : (a) Re_d (b) f

(c) Use Reynold's analogy i.e. $St = \frac{Nu}{Re \cdot Pr} = \frac{f}{8}$, find Nu

(d) $\dot{m}$ (e) $Q = \dot{m} C_p \cdot \Delta T$

(f) $Q = h \cdot A \times \log \text{ mean temperature.}$

$$Re = \frac{V D}{\nu} = \frac{30 \times 0.02}{3.591 \times 10^{-5}} = 1.6708 \times 10^4$$

$$f = 0.3165 (Re)^{-0.25} = 0.3165 (1.6708 \times 10^4)^{-0.25} = 0.02784$$

Reynolds analogy : $St = \frac{Nu}{Re \cdot Pr} = \frac{f}{8}$

$$\text{i.e. } \frac{Nu}{(1.6708 \times 10^4) \times 0.681} = \frac{0.02784}{8}$$

$$Nu = 39.594$$

But, $Nu = \frac{h \cdot D}{k} ;$

$$39.594 = \frac{h \times 0.02}{0.0394}$$

$$h = 78 \text{ W/m}^2 \text{ K}$$

$$\text{Mass flow rate, } \dot{m} = \rho \cdot A \cdot V = \rho \times \frac{\pi}{4} D^2 \times V$$

$$= 0.73 \times \frac{\pi}{4} (0.02)^2 \times 30$$

$$= 0.00688 \text{ kg/s}$$

Rate of heat absorbed by air,

$$Q = \dot{m} C_p \times (T_o - T_i) = 0.00688 \times 1027 \times (270 - 15) = 1801.8 \text{ W}$$

Log mean temperature,

$$\Delta T_{m1} = \frac{T_o - T_i}{\ln \left[\frac{(T_s - T_i)}{(T_s - T_o)} \right]}$$

$$\Delta T_{m1} = \frac{270 - 10}{\ln \left[\frac{(280 - 10)}{(280 - 270)} \right]}$$

$$= 78.89^\circ\text{C}$$

$$\therefore Q = h \cdot A \cdot \Delta T_m$$

$$1801.8 = 78 \times A \times 78.89; \quad A = 0.2928 \text{ m}^2$$

$$A = \pi D L; \quad 0.2928 = \pi \times 0.02 \times L$$

$$L = 4.66 \text{ m}$$

...Ans.

Ex. 6.3.18 : For forced convection from a flat plate, the local Nusselt number is given by the following correlations.

Laminar : $Nu_x = 0.331 Re_x^{0.5} Pr^{0.33}$

Turbulent : $Nu_x = 0.0288 Re_x^{0.8} Pr^{0.33}$

The fins on an air cooled motor cycle engine may be considered as individual flat plates of length $L = 0.2 \text{ m}$ owing to disturbances in the free stream, transition occurs at $Re_{trans} = 2 \times 10^5$. Calculate for a road speed of 140 kmph; the average heat transfer coefficient from the fin surface allowing for separate laminar and turbulent section and compare this result obtained assuming purely turbulent flow.

Assume : $\rho = 1.1 \text{ kg/m}^3, \mu = 1.7 \times 10^{-5} \text{ kg/ms},$

$$k = 0.026 \text{ W/mK}, \quad Pr = 0.7.$$

Soln. :

Given : $L = 0.2 \text{ m}, \quad Re_{trans} = 2 \times 10^5,$

Velocity of air, $V = 140 \text{ kmph} = \frac{140 \times 1000}{3600} = 38.8 \text{ m/sec}$

(It is assumed that fin is stationary and air moving)

Find :

(a) Re to check for laminar / turbulent flow.

(b) Nu

(c) h using Nu

$$\text{Reynold's number, } Re = \frac{V \cdot L \cdot \rho}{\mu} = \frac{38.8 \times 0.2 \times 1.1}{1.7 \times 10^{-5}} = 5.02117 \times 10^5$$

Since $Re > 5 \times 10^5$, flow is turbulent :

$$Nu = 0.0288 (Re)^{0.8} (Pr)^{0.33}$$

$$= 0.0288 (5.02117 \times 10^5)^{0.8} (0.7)^{0.33} = 930.9$$

$$Nu = \frac{h \cdot L}{k} ;$$

$$930.9 = \frac{h \times 0.2}{0.026}$$

$$\therefore h = 121 \text{ W/m}^2\text{K (For turbulent flow) ...Ans.}$$

If flow is assumed laminar with $Re = 2 \times 10^5$ where transition occurs.

$$Nu_1 = 0.331 \cdot Re_x^{0.5} Pr^{0.33}$$

$$= 0.331 (2 \times 10^5)^{0.5} (0.7)^{0.33} = 131.59$$

$$Nu_1 = \frac{h_1 \cdot L}{k} ;$$

$$131.59 = \frac{h_1 \times 0.2}{0.026} ;$$

$$h_1 = 17.1 \text{ W/m}^2\text{K}$$

$$\text{Average heat transfer coefficient} = \frac{h + h_1}{2} = \frac{121 + 17.1}{2}$$

$$= 69 \text{ W/m}^2\text{K} \quad \dots\text{Ans.}$$

Ex. 6.3.19 : Air at 27°C blows along one side of a horizontal brick slab along its 15 m long length. The slab is at 55°C initially. The velocity of air is 4.5 m/s.

Specific heat of slab = 0.84 kJ/kg K ,

Density of slab = 1600 kg/m^3 , volume of slab = 1.5 m^3 .

Properties of air :

$\rho = 1.128 \text{ kg/m}^3$, $\mu = 19.1 \times 10^{-6} \text{ kg/ms}$, $k = 0.0276 \text{ W/m-K}$, $Pr = 0.699$

$Nu = 0.664 Re^{0.5} Pr^{0.33}$ for $Re < 5 \times 10^5$

$Nu = 0.037 Re^{0.8} Pr^{0.33}$ for $Re > 5 \times 10^5$

For unit width of the slab, calculate,

(i) The rate of convective heat loss from the slab,

(ii) Rate of radiative heat loss from slab

(iii) Initial rate of cooling of the slab per hour.

Take emissivity of slab as 0.9.

Soln. :

Given : $T_\infty = 27^\circ\text{C}$, $T_s = 55^\circ\text{C}$, $V = 4.5 \text{ m/sec}$,

$L = 15 \text{ m}$, $W = 1 \text{ m}$.

Find :

(a) Re to check whether flow is laminar or turbulent.

(b) Nu from co-relation

(c) $h = \frac{Nu \cdot k}{L}$ (d) Q_{conv}

(e) $Q_{radiation}$ (f) $Q = Q_c + Q_r$

$$Re = \frac{\rho VL}{\mu} = \frac{1.128 \times 4.5 \times 15}{19.1 \times 10^{-6}} = 3.9863 \times 10^6$$

$\therefore Re > 5 \times 10^5$, flow is turbulent. Hence,

$$Nu = 0.037 Re^{0.8} Pr^{0.33}$$

$$= 0.037 (3.9863 \times 10^6)^{0.8} (0.699)^{0.33}$$

$$= 6271.08$$

$$Nu = \frac{hL}{k} ; h = \frac{Nu \cdot k}{L} = \frac{6271.08 \times 0.0276}{15}$$

$$= 11.538 \text{ W/m}^2\text{K}$$

(i) The rate of convective heat loss from the slab, Q_c :

$$Q_c = h A (T_s - T_\infty)$$

$$= 11.538 \times (15 \times 1) \times (55 - 27)$$

$$\therefore Q_c = 4845 \text{ W}$$

(ii) Rate of radiative heat loss, Q_r :

$$Q_r = \sigma A \epsilon (T_s^4 - T_\infty^4)$$

$$\therefore Q_r = 5.67 \times 10^{-8} \times (15 \times 1) \times 0.9 [(55 + 273)^4 - (27 + 273)^4]$$

$$\therefore Q_r = 2659.4 \text{ W.}$$

(iii) Initial rate of cooling :

$$Q = Q_c + Q_r = 4845 + 2659.4 = 7504.4 \text{ J/sec.}$$

$$Q = -m C_p \frac{dT}{dt} ; \frac{dT}{dt} \text{ is rate of cooling}$$

$$m = \rho V = 1600 \times 1.5 \quad C_p = 840$$

$$\frac{dT}{dt} = \frac{Q}{m C_p} = \frac{7504.4 \times 3600}{1600 \times 1.5 \times 840} = 13.4^\circ\text{C/hr}$$

...Ans.

Ex. 6.3.20 : Oil at 150°C flows slowly through a long, thin-walled pipe of 30-mm inner diameter. The pipe is suspended in a room for which the air temperature is 20°C and the convection coefficient at the outer tube surface is $11 \text{ W/m}^2\text{K}$. Estimate the heat loss per unit length of tube. Assume fully developed flow and negligible thermal resistance of tube wall. For unused engine oil at 423 K ; $k = 0.133 \text{ W/m K}$.

Soln. :

Heat loss from oil at 150°C to air at 20°C is given as;

$$Q = U A \Delta T$$

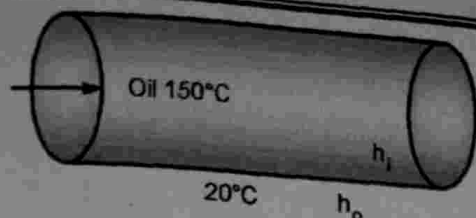


Fig. P. 6.3.20

We know that

$$\frac{1}{U} = \frac{1}{h_i} + \frac{1}{h_o} \text{ neglecting}$$

resistance of cylinder wall

As $h_o = 11 \text{ W/m}^2 \text{ K}$ is given, h_i is required to be determined.

Since oil is flowing slowly but fully developed, so it is laminar flow.

For laminar flow,

$$Nu = 3.66$$

from data book

$$\text{or } \frac{h_i d}{k} = 3.66$$

$$\therefore h_i = 3.66 \times \frac{k}{D} = \frac{3.66 \times 0.133}{0.03}$$

$$= 16.226 \text{ W/m}^2 \text{ K}$$

$$\therefore \frac{1}{U} = \frac{1}{16.226} + \frac{1}{11} = 0.0616 + 0.091$$

$$\therefore U = 6.56 \text{ W/m}^2 \text{ K}$$

Hence heat flow rate per unit length of pipe

$$Q = U A \Delta T = 6.56 \times \pi \times 0.03 \times 1 \times (150 - 20) \\ = 80.33 \text{ W} \quad \dots \text{Ans.}$$

Ex. 6.3.21 : In a tube of 5 cm diameter, the air flows at 24°C and at a velocity of 25 m/s. The surface of the tube is 130°C . Find the heat flow rate through the tube per meter length.

In case the tube is replaced by a duct of square cross-section of 5 cm side with all other data remaining the same, find the heat transfer rates through the duct per meter length.

Assume : $Nu_D = 0.027 Re_D^{0.8} Pr^{0.334}$ for circular tube
 $Nu_D = 0.102 Re_D^{0.68} Pr^{0.334}$ for square duct

Properties of air at 77°C are :

$$\rho = 1.02 \text{ kg/m}^3, \quad \nu = 20.92 \times 10^{-6} \text{ m}^2/\text{s},$$

$$Pr = 0.7, \quad k = 0.03 \text{ W/mK}$$

Compare above heat transfer rates and comment on the results.

Soln. :

Given : $D = 5 \text{ cm} = 0.05 \text{ m}$, $T_\infty = 24^\circ\text{C}$,
 $T_w = 130^\circ\text{C}$, $V = 25 \text{ m/s}$, $L = 1 \text{ m}$,

(1) To find heat transfer rate/m length of tube, Q_1 :

Find :

$$(a) Re_D \quad (b) Nu_D$$

$$(c) h = h_i \quad (d) Q_1$$

$$Re_D = \frac{V \cdot D}{\nu} = \frac{25 \times 0.05}{20.92 \times 10^{-6}} = 5.9751 \times 10^4$$

$$Nu_D = 0.027 Re_D^{0.8} Pr^{0.334} \\ = 0.027 (5.9751 \times 10^4)^{0.8} \times (0.7)^{0.334} = 158.75$$

$$\text{But, } Nu_D = \frac{h_i \cdot D}{k} : 158.75 = \frac{h_i \times 0.05}{0.03}$$

$$h_i = 95.25 \text{ W/m}^2 \text{ K}; A_1 = \pi D L$$

$$Q_1 = h_i \cdot A_1 (T_w - T_\infty) = h_i \times \pi D L \times (T_w - T_\infty)$$

$$Q_1 = 95.25 \times \pi \times 0.05 \times 1 \times (130 - 24)$$

$$= 1585.9 \text{ W}$$

...Ans.

(2) To find heat transfer rate/m length in duct of square cross-section

$a = b = 5 \text{ cm} = 0.05 \text{ m}$, Q_2 :

Find : (a) D_h (b) Re (c) Nu (d) $h = h_2$ (e) Q_2

Hydraulic diameter,

$$D_h = \frac{4 A_c}{P} = \frac{4 (a \times b)}{2 (a + b)} = \frac{4 \times 0.05 \times 0.05}{2 (0.05 + 0.05)}$$

$$= 0.05 \text{ m}$$

$$Re = \frac{V \cdot D_h}{\nu} = \frac{25 \times 0.05}{20.92 \times 10^{-6}} = 5.9751 \times 10^4$$

$$Nu = 0.102 (Re_D)^{0.68} (Pr)^{0.334} \\ = 0.102 (5.9751 \times 10^4)^{0.68} \times (0.7)^{0.334} \\ = 160.24$$

$$\text{But, } Nu = \frac{h_2 \times D_h}{k} ; 160.24 = \frac{h_2 \times 0.05}{0.03}$$

$$h_2 = 96.15 \text{ W/m}^2 \text{ K};$$

$$A_2 = P \cdot L = 2 (a + b) L = 4b \cdot L$$

$$Q_2 = h_2 \cdot A_2 (T_w - T_\infty)$$

$$= h_2 \times (4b \times L) \times (T_w - T_\infty)$$

$$Q_2 = 96.15 \times (4 \times 0.05 \times 1) \times (130 - 24)$$

$$= 2038.4 \text{ W/m}^2 \text{ K}$$

...Ans.

More heat transfer due to more surface area in case of duct.

Ex. 6.3.22 : Air at 20°C flows over a plate 0.5 m wide $\times$ 1 m long at a velocity of 3.2 m/s. The temperature of the plate is maintained at 120°C . Find :

(i) Heat loss per min. from the plate if the air flows parallel to 1.0 m side.



- (ii) What shall be the heat loss per min. from the plate if the air flows parallel to 0.5 m side with all quantities remaining the same.

Use correlation $Nu = 0.332 Re^{1/2} Pr^{1/3}$

Take properties of air at mean temperature.

$$T_m = \frac{20 + 120}{2} = 70^\circ\text{C as :}$$

$$v = 20.02 \times 10^{-6} \text{ m}^2/\text{s}, C_p = 1.009 \text{ kJ/kg K},$$

$$k = 0.02964 \text{ W/mK}$$

$$\rho = 1.029 \text{ kg/m}^3,$$

Soln. :

Given : $T_\infty = 20^\circ\text{C}, W = 0.5 \text{ m},$
 $L = 1 \text{ m}, V = 3.2 \text{ m/s}, T_s = 20^\circ\text{C}.$

- (i) Heat loss from plate if air flows parallel to 1m side, Q_1 :

Find :

(a) Re to check whether flow is laminar or turbulent.

(b) Pr (c) Nu (d) h (e) Q.

$$Re = \frac{V \cdot L}{v} = \frac{3.2 \times 1}{20.02 \times 10^{-6}} = 1.598 \times 10^5$$

$$Re < 3 \times 10^5, \text{ flow is laminar.}$$

$$Pr = \frac{\mu \cdot C_p}{k} = \frac{(v \cdot \rho) C_p}{k}$$

$$= \frac{(20.02 \times 10^{-6} \times 1.029) (1.009 \times 10^3)}{0.02964}$$

$$= 0.7013$$

$$Nu = 0.332 (Re)^{1/2} (Pr)^{1/3}$$

$$= 0.332 (1.598 \times 10^5)^{1/2} \times (0.7013)^{1/3}$$

$$= 117.9$$

$$Nu = \frac{h \cdot L}{k} ; 117.9 = \frac{h \times 1}{0.02964} ;$$

$$h = 3.495 \text{ W/m}^2 \text{ K}$$

$$Q = h \times (2A) \times (T_s - T_\infty)$$

$$= h \times (2 \times L \times W) \times (T_s - T_\infty)$$

$$= 3.495 \times 2 \times 1 \times 0.5 \times (120 - 20)$$

$$= 349.45 \text{ W} \quad \dots\text{Ans.}$$

- (ii) Heat loss from plate if air flows parallel to 0.5m side, Q_2 :

$$L_1 = 0.5 \text{ m}$$

$$\therefore Re = \frac{V \cdot L}{v} = \frac{3.2 \times 0.5}{20.02 \times 10^{-6}} = 0.7921 \times 10^5$$

$$Re < 3 \times 10^5, \text{ flow is laminar.}$$

$$Pr = \frac{\mu \cdot C_p}{k} = 0.7013 \text{ remains the same.}$$

$$Nu = 0.332 (Re)^{1/2} (Pr)^{1/3}$$

$$= 0.332 \times (0.7921 \times 10^5)^{1/2} \times (0.7013)^{1/3}$$

$$= 83.02$$

But, $Nu = \frac{h_1 \cdot L_1}{k} ;$

$$83.02 = \frac{h_1 \times 0.5}{0.02964} ; h_1 = 4.921 \text{ W/m}^2 \text{ K}$$

$$Q_2 = h_1 \times 2A \times (T_s - T_\infty) = h_1 \times 2 (L \times W)$$

$$\times (T_s - T_\infty)$$

$$= 4.921 \times 2 \times (1 \times 0.5) \times (120 - 20)$$

$$= 492.1 \text{ W} \quad \dots\text{Ans.}$$

Ex. 6.3.23 : A 200 W bulb of spherical shape of 7 cm diameter is subjected to flow of air at 30°C . The velocity of air is 0.4 m/s. The surface temperature of the bulb is 120°C . Calculate the rate of heat transfer by convection from bulb surface. At mean temperature, $T_m = 75^\circ\text{C}$ the properties of air are :

$$v = 2.06 \times 10^{-6} \text{ m}^2/\text{s}, k = 0.03 \text{ W/mK}, Pr = 0.7,$$

$$\text{Assume, } Nu = 0.365 Re^{0.6} Pr^{0.33}.$$

Soln. :

Given : $D = 7 \text{ cm} = 0.07 \text{ m}, T_\infty = 30^\circ\text{C},$
 $V = 0.4 \text{ m/s}, T_s = 120^\circ\text{C}.$

Find : (a) Re (b) Using Nu, find 'h'

(c) Q $Re = \frac{V \cdot D}{v} = \frac{0.4 \times 0.07}{2.06 \times 10^{-6}}$

$$= 1.3592 \times 10^4$$

$$\frac{h \cdot D}{k} = Nu = 0.365 Re^{0.6} Pr^{0.33}$$

$$\frac{h \times 0.07}{0.03} = 0.365 (1.3592 \times 10^4)^{0.6} (0.7)^{0.33}$$

$$h = 41.99 \text{ W/m}^2 \text{ K}$$

Heat transfer area for sphere,

$$A = 4\pi R^2 = 4\pi \left(\frac{0.07}{2}\right)^2 = 1.5394 \times 10^{-2} \text{ m}^2$$

$$Q = h \cdot A \cdot (T_s - T_\infty)$$

$$= 41.99 \times 1.5394 \times 10^{-2} \times (120 - 30)$$

$$= 58.175 \text{ W} \quad \dots\text{Ans.}$$

Ex. 6.3.24 : Air at 20°C and velocity 3 m/s flows over a plate of thickness 20 mm. The length and width of the plate are 1m $\times$ 0.5 m. Top surface of plate is maintained at 100°C . Thermal conductivity of plate is 30 W/mK.

Find the heat lost by plate by convection at top surface and the bottom surface temperature of plate under steady state conditions.

Assume the properties of air at mean temperature as :

$$\rho = 1.16 \text{ kg/m}^3, C_p = 1.005 \text{ kJ/kg K}, k_f = 0.02864 \text{ W/mK}$$

$$Pr = 0.7 \text{ and } v = 19 \times 10^{-6} \text{ m}^2/\text{s}.$$

Soln. :

Given : $T_{\infty} = 20^{\circ}\text{C}$, $V = 3 \text{ m/s}$, thickness,
 $\Delta x = 20 \text{ mm} = 0.02 \text{ m}$, $L = 1 \text{ m}$,
 $W = 0.5 \text{ m}$, $T_w = T_s = 100^{\circ}\text{C}$,
 $k = 30 \text{ W/mK}$.

(1) Heat lost by convection from top surface, Q :

Find : (a) Re to check whether flow is laminar or turbulent.

(b) Find Nu and h_x (c) Q .

$$Re = \frac{V \cdot L}{\nu} = \frac{3 \times 1}{19 \times 10^{-6}} = 1.57895 \times 10^5$$

$\therefore Re < 3 \times 10^5$, flow is laminar.

$$Nu = 0.664 (Re)^{1/2} \cdot (Pr)^{1/3}$$

$$= 0.664 (1.57895 \times 10^5)^{1/2} \times (0.7)^{1/3} = 234.3$$

But, $Nu = \frac{h \cdot L}{k_f}$;

$$234.3 = \frac{h \times 1}{0.02894} ; h = 6.78 \text{ W/m}^2 \text{ K}$$

$$Q = h \cdot A \times (T_s - T_{\infty})$$

$$= 6.78 \times (1 \times 0.5) (100 - 20)$$

$$= 271.2 \text{ W} \quad \dots \text{Ans.}$$

(2) Bottom surface temperature of plate, (T_{s1}) :

Heat lost by plate, Q as calculated above must be conducted from bottom surface so as to maintain the top surface temperature of plate. Therefore,

$$Q = \frac{k \cdot A \cdot (T_{s1} - T_s)}{\Delta x} ;$$

$$271.2 = \frac{30 \times (1 \times 0.5) (T_{s1} - 100)}{0.03}$$

$$T_{s1} = 100.5424^{\circ}\text{C} \quad \dots \text{Ans.}$$

Ex. 6.3.25 : In a certain glass making process, a square plate of glass 1 m^2 area and 3 mm thick heated uniformly to 100°C is cooled by air at 20°C flowing over both sides parallel to the plate at 3 m/s . Calculate the initial rate of cooling the plate. Neglect temperature gradient in the glass plate and consider only forced convection.

Take for glass :

$$\rho_g = 2500 \text{ kg/m}^3 \text{ and } C_{pg} = 0.67 \text{ kJ/kg K}$$

Take the following properties of air :

$$\rho_a = 1.076 \text{ kg/m}^3 ;$$

$$C_{pa} = 1008 \text{ J/kg K},$$

$$k = 0.0286 \text{ W/m}^{\circ}\text{C} \text{ and } \mu = 19.8 \times 10^{-6} \text{ N/m}^2.$$

Soln. :

Given : $A = 1 \text{ m}^2$, $T_s = 100^{\circ}\text{C}$, $T_{\infty} = 20^{\circ}\text{C}$,
 $V = 3 \text{ m/s}$, $L = W$ (square plate),
 $\therefore L \times W = 1$ i.e. $L = W = 1 \text{ m}$.

Find :

(a) Re to check whether the flow is laminar and turbulent.

(b) Pr

(c) Nu using co-relations.

(d) 'h'

(e) Q

(f) Equate $Q = m C_{pg} \Delta T_g$.

Initial rate of cooling (dT/dt) :

The average heat transfer co-efficient for the air flow parallel to the plate is given by

(at $x = L = 1 \text{ m}$)

$$Re_L = \frac{\rho_a \cdot V \cdot L}{\mu} = \frac{1.076 \times 3 \times 1}{19.8 \times 10^{-6}} = 163030$$

$$Pr = \frac{\mu \cdot C_{pa}}{k} = \frac{(19.8 \times 10^{-6}) 1008}{0.0286} = 0.698$$

$$Nu = 0.664 (Re_L)^{1/2} (Pr)^{1/3}$$

$$= 0.664 (163030)^{1/2} (0.698)^{1/3} = 237.8$$

$$h = \frac{k}{L} \times Nu = \frac{0.0286}{1} \times 237.8$$

$$= 6.802 \text{ W/m}^2 \text{ / K}$$

Heat flow from both sides of the plate.

$$Q = 2 h A \times (T_s - T_{\infty})$$

$$= 2 \times 6.802 \times 1 \times (100 - 20)$$

$$= 1088.32 \text{ W} \quad \dots \text{Ans.}$$

Instantaneous rate of heat loss by the plate, $Q = \text{Rate of change of internal energy.}$

$$Q = m C_{pg} \frac{dT}{dt} = (\rho_g \times A \times \text{thickness}) C_{pg} \frac{dT}{dt}$$

$$1088.32 = \left(2500 \times 1 \times \frac{3}{1000} \right) \times (0.67 \times 10^3) \times \frac{dT}{dt}$$

$$\frac{dT}{dt} = 0.2166^{\circ}\text{C/sec.} \quad \dots \text{Ans.}$$

Ex. 6.3.26 : 3000 kg of water per hour is heated from 30°C to 70°C by pumping it through a heated pipe. Diameter of tube is 25 mm and its surface temperature is 110°C . Estimate the length of the tube and rate of heat transfer from tube to water.

Thermo physical properties of water at 80°C are :

$$\text{Density} = 972 \text{ kg/m}^3$$

$$\text{Dynamic viscosity} = 355 \times 10^{-6} \text{ kg / m-sec.}$$



Thermal conductivity = 0.667 W/mK

Sp. Heat = 4187 J/kgK

You may use following correlation for turbulent flows through pipes.

$$Nu = 0.023 (Re)^{0.8} (Pr)^{0.33}$$

Soln. :

Given : $m = 3000 \text{ kg/hr}$, $D = 0.025 \text{ m}$,

$V = 1.75 \text{ m/s}$, $L = ?$ $Q = ?$

$T_i = 30^\circ\text{C}$, $T_o = 70^\circ\text{C}$, $T_s = 110^\circ\text{C}$

Find :

(a) Find V using $m = \rho AV$

(b) Re to check whether flow is laminar or turbulent.

(c) Nu (d) ' h ' (e) Q .

$$\text{The mean film temperature } T_{mf} = \frac{1}{2} \left[110 + \left(\frac{30 + 70}{2} \right) \right] \\ = 80^\circ\text{C}$$

$$\dot{m} = \rho AV = \frac{3000}{3600} = 972 \times \frac{\pi}{4} (0.025)^2 \times V$$

$$V = 1.75 \text{ m/s}$$

$$\text{Reynolds Number, } Re = \frac{\rho V D}{\mu}$$

$$= \frac{972 \times 1.75 \times 0.025}{355 \times 10^{-6}} = 119788.73$$

$Re > 2000$, hence the flow is turbulent

$$\text{Using correlation: } Nu = 0.023 (Re)^{0.8} (Pr)^{0.4}$$

$$Pr = \frac{\mu C_p}{k} = \frac{355 \times 10^{-6} \times 4187}{0.667} = 2.228$$

$$Nu = 0.023 (119788.73)^{0.8} (2.228)^{0.4} = 366.1$$

$$Nu = \frac{h \cdot D}{k}$$

$$h = \frac{Nu \cdot k}{D} = \frac{366.1 \times 0.667}{0.025} = 9768.96 \text{ W/m}^2 \text{ K}$$

(1) Length of pipe, L :

The rate of heat transfer, $Q = h A (T_s - T_\infty)$

$$= m C_p (T_o - T_i) \text{ (where, } A = \pi D L)$$

$$9768.96 \times (\pi \times 0.025 \times L) (110 - 50) = \frac{3000}{3600} \times 4187 \times (70 - 30)$$

$$L = 3.03 \text{ m} \quad \dots \text{Ans.}$$

(2) Rate of heat transfer :

$$Q = m C_p (T_o - T_i)$$

$$= \frac{3000}{3600} \times 4187 \times (70 - 30)$$

$$L = 139.56 \text{ kJ/s or kW} \quad \dots \text{Ans.}$$

Ex. 6.3.27 : Air flows with a velocity of 0.5 m/s through a rectangular cross-sectioned duct with dimensions 10 cm $\times$ 5 cm and length = 5.67 m. The duct is heated uniformly throughout its length. The duct wall temperature is 20°C higher than the air temperature throughout its length. If the bulk mean temperature of air is 27°C, calculate the rate of heat transfer between duct and air.

Now, air velocity is made = 2 m/s. As a result, duct wall temperature is 10°C higher than air temperature throughout the length of the duct. Find the percentage change in rate of heat transfer between this case and the previous case.

Correlations : $Nu = 4.364$ for laminar flow.

$$= 0.023 Re^{0.8} Pr^{0.4} \text{ turbulent flow.}$$

Take : $\nu = 15.89 \times 10^{-6} \text{ m}^2/\text{s}$ at $T = 300 \text{ K}$,

$$\rho = 1.1614 \text{ kg/m}^3, Pr = 0.707.$$

Soln. :

Given : $V = 0.5 \text{ m/s}$, Cross-section of duct = 10 cm $\times$ 5 cm, = 0.1 m $\times$ 0.05 m

$\therefore$ Perimeter, $P = 2 (0.1 + 0.05) = 0.3 \text{ m}$, $L = 5.67 \text{ m}$,

$T_s - T_\infty = 20^\circ\text{C}$,

Bulk mean temperature, $T_{bm} = 27^\circ\text{C} = 300 \text{ K}$

Surface area = Perimeter, $P \times$ Length,

$$L = 0.3 \times 5.67 = 1.701 \text{ m}^2$$

Find :

(a) D_h (b) Re to check for type of flow.

(c) Nu (d) ' h ' (e) Q .

$$\text{Hydraulic diameter, } D_h = \frac{4A}{P} = \frac{4 \times (0.1)(0.05)}{0.3} \\ = 0.0667 \text{ m}$$

$$\text{Case I : Reynolds Number} = \frac{V \cdot D_h \cdot \rho}{\mu}$$

$$= \frac{V \cdot D_h}{\nu} = \frac{0.5 \times 0.0667}{15.89 \times 10^{-6}} = 2099$$

$$\text{Hence flow is laminar } \therefore Nu = 4.364 = \frac{h \cdot D_h}{k}$$

$$\therefore h = \frac{4.364 \times k}{D_h} = \frac{4.364 \times 0.0263}{0.0667}$$

$$= 1.72 \text{ W/m}^2 \text{ K}$$

$$Q = h \cdot A_s (T_s - T_\infty)$$

$$= 1.72 \times 1.701 \times 20 = 58.5 \text{ W} \quad \dots \text{Ans.}$$

Case II : When $V = 2 \text{ m/s}$, $T_s - T_\infty = 10^\circ\text{C}$

$$\text{Reynolds Number} = \frac{V \cdot D_h}{\nu} = \frac{2 \times 0.0667}{15.89 \times 10^{-6}} \\ = 8395.2$$

Hence flow is turbulent

$$Nu = 0.023 \times Re^{0.8} Pr^{0.4}$$

$$= 0.023 (8395.2)^{0.8} (0.707)^{0.4} = 27.58$$

$$Nu = \frac{h \cdot D_h}{k}; \quad h = \frac{Nu \cdot k}{D_h} = \frac{27.58 \times 0.0263}{0.06667} = 10.87 \text{ W/mK}$$

$$Q_1 = h \cdot A (T_s - T_\infty) = 10.87 \times 1.701 \times 10 = 184.8 \text{ W} \quad \dots \text{Ans.}$$

percentage increase in heat transfer rate :

$$= \frac{Q_1 - Q}{Q} \times 100\% = \frac{(184.8 - 58.5)}{58.5} \times 100 = 215.9\% \quad \dots \text{Ans.}$$

Ex. 6.3.28 : A refrigerated truck is travelling at 90 kmph on a desert highway where the air temperature is 60°C. The body of the truck may be idealised as a rectangular box, 3.2 m wide, 2.1 m high and 6.0 m long, at a surface temperature of 10°C. Assume that the heat transfer from the front and back of the truck may be neglected, that the stream does not separate from the surface and that the boundary layer is turbulent over the whole surface. Calculate (a) the rate of heat transfer to the four surfaces, (b) the required tonnage of the refrigeration unit and (c) the power required to overcome the resistance acting on the four surfaces.

Properties of air at 35°C are : $\rho = 1.147 \text{ kg/m}^3$, $k = 0.0271 \text{ W/mK}$,

$$\nu = 16.48 \times 10^{-6} \text{ m}^2/\text{s},$$

$$\text{And } C_p = 1.005 \text{ kJ/kg K.}$$

$$\text{Assume, } C_f = 0.072 \text{ Re}^{-0.2}.$$

Soln. :

$$\text{Given : } V = 90 \text{ kmph} = \frac{90 \times 1000}{3600} = 25 \text{ m/s},$$

$$T_s = 60^\circ\text{C},$$

$$W = 3.2 \text{ m}, \quad H = 2.1 \text{ m},$$

$$L = 6 \text{ m}, \quad T_\infty = 10^\circ\text{C}.$$

$$Re_L = \frac{VL}{\nu} = \frac{25 \times 6}{16.48 \times 10^{-6}} = 9.1 \times 10^6 \quad (\text{Flow is turbulent})$$

$$Pr = \frac{\nu \rho C_p}{k} = \frac{16.48 \times 10^{-6} \times 1.147 \times (1.005 \times 10^3)}{0.0271} = 0.7$$

For turbulent flow, the Nusselt Number is given as :

$$Nu = 0.036 (Re_L)^{0.8} (Pr)^{1/3} = 0.036 (9.1 \times 10^6)^{0.8} (0.7)^{1/3} = 11800.7$$

$$\text{But, } Nu = \frac{h \cdot L}{k}; \quad h = \frac{Nu \cdot k}{L} = \frac{11800.7 \times 0.071}{6} = 139.64 \text{ W/m}^2 \text{ K}$$

(a) **Rate of heat transfer, Q :**

Surface area for heat transfer,

$$A = 2(W + H)L = 2(3.2 + 2.1)6 = 63.6 \text{ m}^2$$

$$Q = h \cdot A (T_s - T_\infty) = 139.64 \times 63.6 (60 - 10)$$

$$= 444.055 \times 10^3 \text{ W}$$

$$= 444.055 \text{ kW} \quad \dots \text{Ans.}$$

(b) **Required tonnage of refrigeration, R_E :**

$$(\because 1 \text{ TR} = 3.517 \text{ kW})$$

$$R_E = \frac{Q}{3.517} = \frac{444.055}{3.517} = 126.26 \text{ TR} \quad \dots \text{Ans.}$$

(c) **Power required to overcome resistance acting on four faces, P :**

$$\text{Friction coefficient, } C_f = 0.072 (Re_L)^{-0.2} = 0.072 (9.1 \times 10^6)^{-0.2} = 0.00292$$

$$\text{Drag force, } F_D = C_f \cdot \frac{\rho \cdot A \cdot V^2}{2} = 0.00292 \times \frac{1.147 \times 63.6 \times 25^2}{2} = 66.566 \text{ N}$$

$$\text{Power required, } P = F_D \cdot V = 66.566 \times 25 = 1664.2 \text{ Nm/s or W} \quad \dots \text{Ans.}$$

Ex. 6.3.29 : In a power plant, feed water is flowing through a rectangular duct 8 cm $\times$ 4 cm and the wall temperature is maintained at 170°C throughout. The feed water flows at the rate of 300 kg/min, enters at a temperature of 20°C and is heated to 150°C. Compare the heat transfer coefficients using.

(i) Dittus - Boelter Equation. (ii) Sieder - Tate Equation. Also estimate the required length of the duct. Properties of water at 85°C are :

$$Pr = 1.64, \mu = 265 \times 10^{-6} \text{ kg/ms}, C_p = 4.226 \text{ kJ/kg K.}$$

$$k = 683 \times 10^{-6} \text{ kW/mK}, \mu_w \text{ at } 170^\circ\text{C} = 158 \times 10^{-6} \text{ kg/ms.}$$

Soln. : Refer Fig. P. 6.3.29

Given :

$$\dot{m} = 300 \text{ kg/min} = \frac{300}{60} = 5 \text{ kg/s}, \quad T_s = 170^\circ\text{C},$$

$$T_i = 20^\circ\text{C}, \quad T_o = 150^\circ\text{C},$$

$$A_c = 8 \text{ cm} \times 4 \text{ cm} = 32 \text{ cm}^2 = 32 \times 10^{-4} \text{ m}^2$$

$$P = 2(a + b) = 2(0.08 + 0.04) = 0.24 \text{ m}$$

$$\text{Surface area of duct, } A_s = P \cdot L = 0.24 \text{ L}$$

Find :

(a) Hydraulic diameter, D_h

(b) Log mean temperature.

(c) Re_d

(d) Nu_d

(e) h .

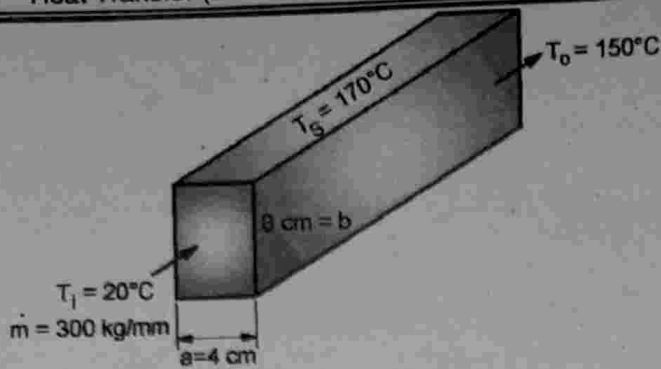


Fig. P. 6.3.29

Hydraulic diameter of duct,

$$D_h = \frac{4 A_c}{P} = \frac{4 \times 32 \times 10^{-4}}{0.24} = 0.0533 \text{ m}$$

Mean temperature, $\Delta T_{lm} = \frac{T_o - T_i}{\ln \left(\frac{T_s - T_i}{T_s - T_o} \right)}$

$$= \frac{150 - 20}{\ln \left(\frac{170 - 20}{170 - 150} \right)} = 64.52^\circ\text{C}$$

$$\begin{aligned} Re_D &= \frac{\rho V D}{\mu} = \frac{\dot{m} D_h}{A \cdot \mu} \\ &= \frac{300}{60} \times \frac{0.0533}{(32 \times 10^{-4}) \times (265 \times 10^{-6})} \\ &= 314260 \end{aligned}$$

(a) Dittus - Boelter Equation :

$$\begin{aligned} Nu_D &= 0.023 (Re_D)^{0.8} (Pr)^{0.4} \\ &= 0.023 (314260)^{0.8} (1.64)^{0.4} = 700.6 \\ h &= \frac{Nu_D \cdot k}{D_h} = \frac{700.6 \times 683 \times 10^{-6}}{0.0533} \\ &= 8.978 \text{ kW/m}^2 \text{ K} \end{aligned}$$

...Ans.

(b) Using Sieder Tate Equation :

$$\begin{aligned} Nu_D &= 0.027 Re_D^{0.8} Pr^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14} \\ Nu &= 0.027 \times (314260)^{0.8} (1.64)^{0.33} \\ &\quad \times \left(\frac{265 \times 10^{-6}}{158 \times 10^{-6}} \right)^{0.14} = 854.2 \\ h &= \frac{Nu_D \cdot k}{D_h} = \frac{854.2 \times (683 \times 10^{-6})}{0.0533} \\ &= 10.94 \text{ kW/m}^2 \text{ K} \end{aligned}$$

...Ans.

(c) Length of the duct, L :

Value of $h = 10.96 \text{ kW/m}^2 \text{ K}$ is more exact since it takes into account the viscous effects with temperature. We shall consider this value of h for calculation of length of duct.

$$\begin{aligned} Q &= \dot{m} C_p (T_o - T_i)_{\text{water}} \\ &= 5 \times 4.226 (150 - 20) = 2747 \text{ kW} \end{aligned}$$

$$Q = h \cdot A \cdot \Delta T_{lm}$$

$$\therefore 2747 = 10.96 \times 2 (0.04 + 0.08) L \times 64.52$$

$$L = 16.186 \text{ m}$$

...Ans.

Ex. 6.3.30 : 4800 kg/hr of water is heated from 30°C to 60°C by passing through a square duct of $30 \text{ mm} \times 30 \text{ mm}$. The duct is heated by condensing steam at 100°C on its outer surface. Find the length of the duct required.

Take properties of water : Density = 995 kg/m^3 ; $\mu = 7.65 \times 10^{-4} \text{ kg/ms}$; $C_p = 4.174 \text{ kJ/kg K}$; $k = 0.623 \text{ W/m}^\circ\text{C}$; Conductivity of duct material = $24 \text{ W/m}^\circ\text{K}$

Use : $Nu = 0.023 Re^{0.8} Pr^{0.4}$ for turbulent flow

$Nu = 4.36$ for laminar flow.

Soln. :

Given :

$$\dot{m} = 4800 \text{ kg/h} = 1.334 \text{ kg/s}$$

$$T_i = 30^\circ\text{C}; T_o = 60^\circ\text{C}; T_s = 100^\circ\text{C}$$

Line of approach :

We know that heat transfer rate

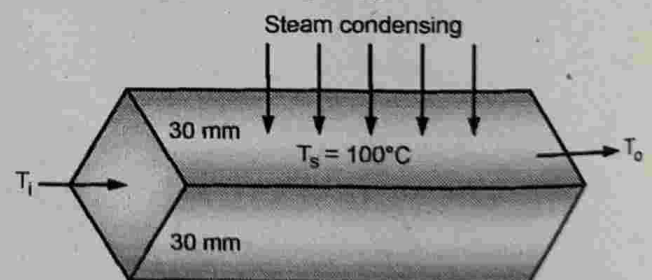


Fig. P. 6.3.30

$$Q = h A \Delta T$$

Area A is outer perimeter of duct multiplied by its length L (which is required to be found out). h can be found out from Nu relations. To obtain appropriate Nu , we should know whether flow is Laminar or turbulent. we should find out Reynold's number. To know this,

$$Re = \frac{\rho V D}{\mu}$$

$$D = D_h = \frac{4A}{P} = \frac{4 \times 0.03 \times 0.03}{4 \times 0.03} = 0.03 \text{ m}$$

$$m = \rho A V$$

$$\therefore \rho V = \frac{m}{A} = \frac{1.334}{0.03 \times 0.03} = 1482.2$$

$$\therefore Re = \frac{1482.2 \times 0.03}{7.65 \times 10^{-4}} = 58,126.4 > 4000$$

Hence flow is turbulent

Therefore we will use

$$Nu = 0.023 Re^{0.8} Pr^{0.4} \text{ relation to find out } h.$$

$$Pr = \frac{\mu C}{k} = \frac{7.6 \times 10^{-4} \times 4174}{0.623} = 5.13$$

$$Nu = \frac{h D_h}{k} = 0.023 \times (58126.4)^{0.8} (5.13)^{0.4}$$

$$h = \frac{0.623 \times 0.023}{0.03} \times (58126.4)^{0.8} (5.13)^{0.4}$$

$$= 5951.7 \text{ W/m}^2 \text{ K}$$

$$Q = h A \Delta T = m C (T_o - T_i)$$

$$= 5951.7 \times 2 (0.03 + 0.03) \times L \times \left[100 - \frac{30 + 60}{2} \right]$$

$$= 1.334 \times 4174 (60 - 30)$$

$$\therefore L = 4.25 \text{ m}$$

...Ans.

Ex. 6.3.31 : Liquid metal at the rate of 5 kg/s flows in a tube of 6 cm diameter. It enters at 400°C into the tube and leaves at 430°C. A constant heat flux is maintained along the tube. The tube surface temperature is 22°C higher than the temperature of liquid metal. Calculate the length of tube required for the purpose. Take properties of liquid metal :

$$\mu = 1.35 \times 10^{-3} \text{ kg/ms}, C_p = 150 \text{ J/kgK}, Pr = 0.011,$$

$$k = 16 \text{ W/mK. Use the following correlation :}$$

$$Nu = 4.82 + 0.0185 (Re \cdot Pr)^{0.83}$$

Soln. :

Given :

$$d = 6 \text{ cm} = 0.06 \text{ m}$$

$$m = 5 \text{ kg/s}, T_i = 400^\circ \text{C}$$

$$T_o = 430^\circ \text{C}; \Delta T = 22^\circ \text{C}$$

$$\mu = 1.35 \times 10^{-3} \text{ kg/ms}, C_p = 150 \text{ J/kg K}$$

$$Pr = 0.011; k = 16 \text{ W/m K}$$

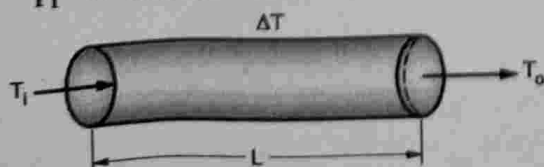


Fig. P. 6.3.31

Line of approach :

Length of pipe can be found out from surface $A = \pi dL$ transferring heat to liquid metal from the expression.

$$Q = h A \Delta T$$

$$Q = m C_p (T_o - T_i);$$

h can be found out from given

$$Nu = 4.82 + 0.0185 (Re \cdot Pr)^{0.83};$$

$$\Delta T = 22^\circ \text{C (given)}$$

$$m = \rho \cdot a \cdot V = \rho \times \frac{\pi}{4} d^2 \times V = 5 \text{ kg/s}$$

$$\therefore V = \frac{1769}{\rho}$$

$$Re = \frac{\rho V d}{\mu} = P \times \frac{1769}{\rho} \times \frac{0.06}{1.35 \times 10^{-3}} = 78635$$

$$\therefore Nu = 4.82 + 0.0185 (78635 \times 0.011)^{0.83} = 9.889$$

$$Nu = \frac{h d}{k} = 9.889 \Rightarrow h = \frac{9.889 \times 16}{0.06}$$

$$= 2637 \text{ W/m}^2 \text{ K}$$

$$Q = h A \Delta T = m C_p (T_o - T_i)$$

$$\therefore 2637 \times \pi \times 0.06 \times L \times 22 = 5 \times 150 (430 - 400)$$

$$\text{Hence } L = 2 \text{ m}$$

...Ans.

Ex. 6.3.32 : Liquid mercury flows at a rate of 1.6 kg/s through a copper tube of 20 mm diameter. The mercury enters the tube at 15°C and leaves at 35°C. calculate the tube length if the tube wall temperature is 50°C. the properties of mercury at 25°C are $\rho = 13582 \text{ kg/m}^3$, $C_p = 140 \text{ J/kg K}$, $k = 8.69 \text{ W/m K}$, $\nu = 1.5 \times 10^{-7} \text{ m}^2/\text{s}$, $Pr = 0.0248$.

$$\text{Use } Nu = 7 + 0.025 (Re \cdot Pr)^{0.8}$$

Soln. :

Given :

$$m = 1.6 \text{ kg/s}; D = 20 \text{ mm} = 0.02 \text{ m}$$

$$T_i = 15^\circ \text{C}; T_e = 35^\circ \text{C}; T_w = 50^\circ \text{C}$$

$$T_m = \frac{15 + 35}{2} = 25^\circ \text{C}$$

$$\rho = 13582 \text{ kg/m}^3;$$

$$C_p = 140 \text{ J/kg K};$$

$$k = 8.69 \text{ W/m K};$$

$$\nu = 1.5 \times 10^{-7} \text{ m}^2/\text{s}; Pr = 0.0248.$$

$$Nu = 7 + 0.025 (Re \cdot Pr)^{0.8Q}$$

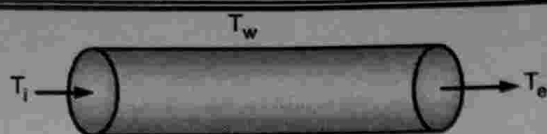


Fig. P. 6.3.32

$$m = \rho AV$$

$$V = \frac{m}{\rho A} = \frac{1.6}{13582 \times \frac{\pi}{4} \times 0.02^2} = 0.375 \text{ m/s}$$

$$Re = \frac{VD}{\nu} = \frac{0.375 \times 0.02}{1.5 \times 10^{-7}} = 50,000$$

$$Nu = 7 + 0.025 (Re \cdot Pr)^{0.8}$$

$$\frac{hD}{k} = 7 + 0.025 (50000 \times 0.0248)^{0.8} ;$$

$$h = \frac{k}{D} [7 + 0.025 (1240)^{0.8}]$$

$$h = \frac{8.69}{0.02} [7 + 0.025 (1240)^{0.8}] = 6282.4 \text{ W/m}^2\text{K}$$

$$Q = h A \Delta T = h \cdot \pi DL (T_w - T_m)$$

$$\text{Heat flow rate, } Q = m C_p \Delta T \\ = 1.6 \times 140 \times (35 - 15) = 4480 \text{ W}$$

$$\text{But, } Q = h \cdot A (T_w - T_m)$$

$$4480 = 6282.4 \times \pi \times 0.02 \times L \times (50 - 25)$$

$$L = 0.454 \text{ m} = 45.4 \text{ cm} \quad \dots \text{Ans.}$$

6.4 Mechanism of Natural Convection

We have so far discussed the heat transfer between the surface and a fluid where the flow of fluid is maintained by a fan, blower, pump or other pressure producing devices.

There is another case of convecting heat transfer in which the convection currents exist due to change in density resulting from temperature gradient between the hot surface and the fluid. Such a heat transfer between the fluid and surface is called by **natural or free convection**.

Examples of free convection are :

1. Cooling of human body largely by natural convection.
2. Flow of hot gases in a chimney of a power plant.
3. Flow of hot air movement along the road in summer hot days.
4. Flow of hot air from the surface of plate or a cylinder or a wall.
5. Cooling of electronic devices by the cool air entering through the slots at the bottom and hot air escaping from slots at the top of cabinet.

6. Internal cooling of turbine blades.
7. Household steam radiators.

In all the above cases of natural convection, the cooling is not affected by the draft of a fan or blower and yet convection currents are set up.

The mechanism of natural convection is as follows.

Consider a hot vertical plate kept in cool air at rest as shown in Fig. 6.4.1.

When the air comes in contact with the hot surface, its molecules in the immediate vicinity of the surface receive heat from the hot surface. This causes the temperature of molecules and its specific volume (i.e. air becomes lighter) to rise. The density of air decrease due to increase in temperature of air adjacent to the surface. Thus, this hot air rises up under the action of buoyancy forces (*Note that the buoyancy force is equal to the weight of molecules displaced, hence it depends on the volume of molecules displaced.*).

The place of the hot air is taken up by heavier molecules (cold air having higher density) which will also rise in similar manner on receiving the energy from hot surface.

This way the natural motion in the fluid molecules will be set in.

The transfer of heat from solid surface to the fluid due to density variation in the fluid caused by the temperature gradient between the surface and the fluid is called **heat transfer by natural convection**.

In the example explained above, the buoyancy force is more than the gravitational force (it depends on the mass of molecules which is constant) due to which motion is set up. However, the body forces need not be due to gravity only e.g. in case of rotating turbine blades, the centrifugal force exists as the body force that drives the fluid motions in presence of density difference that are caused by temperature differences.

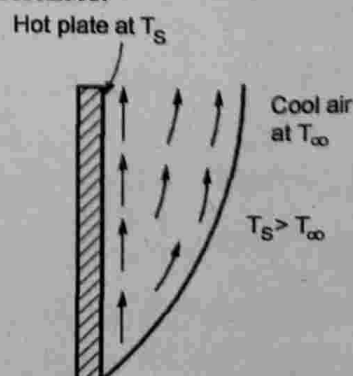


Fig. 6.4.1 : Natural convection-cooling of hot plate

The natural convection currents set in depends upon the shape of object and the temperatures of surface, T_s and fluid, T_∞ . The natural convection currents set up over some surfaces are shown in Fig. 6.4.2.

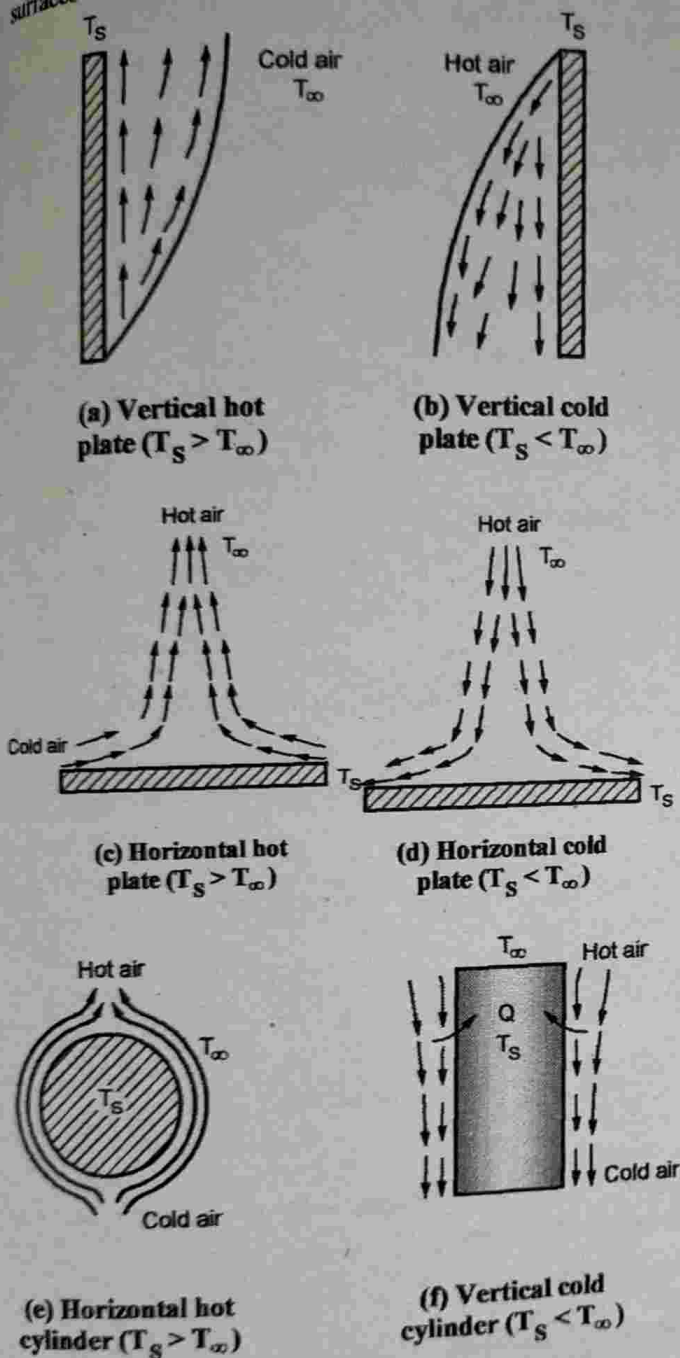


Fig. 6.4.2 : Natural convection currents on various cases

6.5 Coefficient of Volumetric Expansion (β)

The coefficient of volumetric expansion, β is defined as the density of fluid variation per degree change in temperature at constant pressure. Mathematically,

$$\beta = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_{p=\text{constant}} \quad \dots(6.5.1)$$

Negative sign since density decreases with increase in temperature, thus $\left(\frac{\partial \rho}{\partial T} \right)$ is negative.

$$\beta = -\frac{1}{\rho} \times \frac{\Delta \rho}{\Delta T} \text{ at } p = \text{constant}$$

$$\text{i.e. } \Delta \rho = -\rho \cdot \beta \cdot \Delta T \quad \dots(6.5.2)$$

For an ideal gas, $p = \rho R T$. Hence, $\rho = \frac{p}{RT}$

Therefore, $\frac{\partial \rho}{\partial T} = -\frac{1}{T^2} \times \frac{p}{R}$. On substituting the values of ρ and $\frac{\partial \rho}{\partial T}$ in Equation (6.5.1) we get,

$$\begin{aligned} \beta &= -\frac{1}{\left(\frac{p}{RT} \right)} \times \left(-\frac{1}{T^2} \times \frac{p}{R} \right) \\ &= \frac{1}{T} \text{ (per Kelvin)} \quad \dots(6.5.3) \end{aligned}$$

The concept of coefficient of volumetric expansion is important in natural convection problems since natural convection is set up due to density variation caused by temperature gradient.

Higher is the value of β , higher will be the density gradient hence the heat transfer rates.

6.6 Dimensional Analysis of Natural Convection

Consider a hot vertical plate of characteristic length, L having surface temperature T_s which is kept in surrounding air at temperature, T_∞ with negligible velocity as shown in Fig. 6.6.1.

The air gets heated in the vicinity of surface. It causes the density of air to decrease. The hot air rises up due to density difference and the free convection current are set up. The velocity of air in free convection is small. The magnitude of the buoyancy force is equal to the weight of air / fluid displaced by the air. Let ρ_∞ be the bulk density of air and ρ is the density of hot air having the temperature difference ΔT and coefficient of volumetric expansion β . Then, $\rho_\infty = \rho (1 + \beta \cdot T)$ and,

$$\text{Buoyant force / unit volume} = (\rho_\infty - \rho) g = \rho \cdot g \cdot \beta \cdot \Delta T.$$

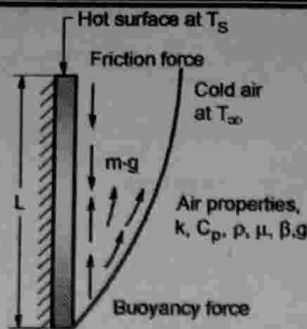


Fig. 6.6.1 : Natural convection in hot vertical plate

Factors affecting heat transfer coefficient, h :

Based on the discussion above and the concepts used in forced convection, we find that the heat transfer coefficient, ' h ' depends on various parameters as follows in natural convection :

$$h = f(\rho \cdot g, \beta, \Delta T, L, \mu, C_p, k) \quad \dots(i)$$

The term $g \cdot \beta \cdot \Delta T$ represents the buoyant force / unit volume. Therefore $(g \cdot \beta \cdot \Delta T)$ in Equation (i) can be combined into one term. Then the convective heat transfer coefficient, h can be written as :

$$h = C (g \cdot \beta \cdot \Delta T)^a \rho^b L^c \mu^d C_p^e k^f \quad \dots(ii)$$

where, C, a, b, c, d, e and f are constants,

The fundamental dimensions of the various variables are as follows :

Variables	Units	Fundamental dimensions
Convective heat transfer coefficient, h	$\frac{W}{m^2 \cdot K}$	$M t^{-3} T^{-1}$
$g \cdot \beta \cdot \Delta T$	$\frac{m}{s^2} \times \frac{1}{K} \times K = \frac{m}{s^2}$	$L t^{-2}$
Density, ρ	$\frac{kg}{m^3}$	$M L^{-3}$
Characteristic length, L	M	L
Dynamic viscosity, μ	$\frac{kg}{m \cdot s}$	$M L^{-1} t^{-1}$
Specific heat, C_p	$\frac{Nm}{kgK} = \frac{kg \cdot m}{s^2 \cdot K} \times \frac{m}{kgK} = \frac{m^2}{s^2 K}$	$L^2 t^{-2} T^{-1}$
Thermal conductivity, k	$\frac{W}{mK} = \frac{Nm}{s \cdot K} \times \frac{1}{mK} = \frac{kg \cdot m}{s \cdot s^2 \cdot K} \times \frac{1}{K}$	$M L t^{-3} T^{-1}$

Substituting the fundamental dimensions in Equation (ii) we get.

$$M \cdot L^0 t^{-3} T^{-1} = C [(L t^{-2})^a \cdot (M L^{-3})^b (L)^c (M L^{-1} t^{-1})^d (L^2 t^{-2} T^{-1})^e (M L t^{-3} \cdot T^{-1})^f]$$

On Equating the powers of M, L, t and T on both sides,

$$\text{Equating powers for } M: 1 = b + d + f \quad \dots(iii)$$

$$\text{Equating powers for } L: 0 = a - 3b + c - d + 2e + f \quad \dots(iv)$$

$$\text{Equating powers for } t: -3 = -2a - d - 2e - 3f \quad \dots(v)$$

$$\text{Equating powers for } T: -1 = -e - f \quad \dots(vi)$$

From Equation (vi),

$$f = 1 - e \quad \dots(vii)$$

From Equation (v),

$$-3 = -2a - d - 2e - 3(1 - e)$$

$$-3 = -2a - d - 2e - 3 + 3e$$

$$-3 = -2a - d + e - 3$$

$$\text{or, } d = -2a + e \quad \dots(viii)$$

From Equation (iii),

$$1 = b + (-2a + e) + (1 - e)$$

$$b = 2a \quad \dots(ix)$$

From Equation (iv),

$$0 = a - 3(2a) + c - (-2a + e) + 2e + (1 - e)$$

$$0 = -3a + c \quad \text{i.e. } c = (3a - 1) \quad \dots(x)$$

On substituting the values of constants in Equation (ii) we get,

$$h = C [(g \cdot \beta \cdot \Delta T)^a (\rho)^{2a} (L)^{3a-1} (\mu)^{-2a+e} \cdot (C_p)^e (k)^{1-e}]$$

$$h = C$$

$$\left[(g \cdot \beta \cdot \Delta T \rho^2 L^3 \mu^{-2})^a \times \frac{1}{L} \times \left(\frac{\mu \cdot C_p}{k} \right)^e \cdot k \right]$$

$$\frac{h \cdot L}{k} = C \left[\left(\frac{g \cdot \beta \cdot \Delta T \rho^2 L^3}{\mu^2} \right)^a \times \left(\frac{\mu \cdot C_p}{k} \right)^e \right] \quad \dots(6.6.1)$$

Equation (6.6.1) contains three dimensionless parameters as follows :

$$1. \text{ Nusselt's number, } Nu = \frac{h \cdot L}{k} \quad \dots(6.6.2)$$

$$2. \text{ Grashoff's number, } Gr = \frac{g \cdot \beta \cdot \Delta T \rho^2 L^3}{\mu^2} = \frac{g \cdot \beta \cdot \Delta T L^3}{\nu^2} \quad \dots(6.6.3)$$

$$\therefore \nu = \left(\frac{\mu}{\rho} \right)$$

$$3. \text{ Prandtl's number, } Pr = \frac{\mu \cdot C_p}{k} \quad \dots(6.6.4)$$

Thus the generalized form of Equation (6.6.1) can be written as :

$$Nu = C \cdot (Gr)^a \cdot (Pr)^e = C (Gr \cdot Pr)^\eta \quad \dots(6.6.5)$$

Equation (6.6.5) becomes the governing Equation for natural convection.

6.7 Correlations Applicable to Natural Convection

Formulae and correlations to be used in natural convection are given in Table 6.7.1

Table 6.7.1 : Formulae and correlation related to natural convection

1. Coefficient of volumetric expansion,

$$\beta = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_{p = \text{constant}}$$

For ideal gas : $\beta = \frac{1}{T}$ (where temperature, T is in Kelvin)

$T = T_m$ = Mean film temperature.

2. Nusselt number, $Nu = \frac{h \cdot L}{k}$ (where L is characteristics length)

3. Grashoff Number, $Gr = \frac{g \cdot \beta \cdot \Delta T \cdot L^3}{\nu^2}$ [where, $\nu = \frac{\mu}{\rho}$, $\Delta T = (T_s - T_\infty)$ or $(T_\infty - T_s)$]

4. Prandtl number, $Pr = \frac{\mu \cdot C_p}{k}$

5. $Nu = C (Gr)^a \cdot (Pr)^b = C (Gr Pr)^n$

6. Rayleigh number, $Ra = Gr \cdot Pr = \frac{g \cdot \beta \cdot L^3 \cdot \Delta T}{\nu \cdot \alpha}$

Correlations :

Sr. No	Geometry	Correlation for Nu	Condition for applicability $Ra = Gr Pr$	Characteristic length, $L = L_c$
1.	Vertical plates Vertical cylinder	$Nu = 0.59 (Gr \cdot Pr)^{0.25}$ $Nu = 0.13 (Gr \cdot Pr)^{1/3}$	$10^4 < Gr Pr < 10^9$ $10^9 < Gr Pr < 10^{12}$	Vertical length
2.	Horizontal plates (facing up)	$Nu = 0.54 (Gr \cdot Pr)^{0.25}$ $Nu = 0.15 (Gr \cdot Pr)^{1/3}$	$10^5 < Gr Pr < 2 \times 10^7$ $2 \times 10^7 < Gr Pr < 3 \times 10^{10}$	$L_c = \frac{A}{P}$
3.	Horizontal plates (facing down)	$Nu = 0.27 (Gr \cdot Pr)^{0.25}$	$3 \times 10^5 < Gr Pr < 3 \times 10^{16}$	$L_c = \frac{A}{P}$
4.	Horizontal cylinder	$Nu = 0.53 (Gr \cdot Pr)^{0.25}$ $Nu = 0.13 (Gr \cdot Pr)^{1/3}$	$10^4 < Gr Pr < 10^9$ $10^9 < Gr Pr < 10^{12}$	Diameter, D
5.	Sphere	$Nu = 2 + 0.43 (Gr \cdot Pr)^{0.25}$	$1 < Gr Pr < 10^5$	Diameter, D
6.	Square / Circular plates (facing up)	$Nu = 0.54 (Gr \cdot Pr)^{0.25}$ $Nu = 0.15 (Gr \cdot Pr)^{1/3}$	$10^5 < Gr Pr < 2 \times 10^7$ $2 \times 10^7 < Gr Pr < 2 \times 10^{10}$	$L_c = \frac{A}{P}$
7.	Square / Circular plate (from lower surface)	$Nu = 0.27 (Gr \cdot Pr)^{0.25}$	$3 \times 10^5 < Gr Pr < 3 \times 10^{16}$	$L_c = \frac{A}{P}$



Ex. 6.7.1: Estimate the heat transfer rate from a 100 W incandescent bulb at 140°C to an ambient at 24°C.

Approximate the bulb as 60 cm diameter sphere. Calculate the percentage of power lost by natural convection. Use following correlation and air properties:

$$Nu = 0.60 (Gr Pr)^{1/4}$$

The properties of air at 82°C are $\nu = 21.46 \times 10^{-6} \text{ m}^2/\text{s}$, $K = 30.38 \times 10^{-3} \text{ W/m}\cdot\text{K}$, $Pr = 0.699$

Soln.:

Given: $D = 60 \text{ cm} = 0.6 \text{ m}$; $T_s = 140^\circ\text{C} = 413 \text{ K}$;
 $T_\infty = 24^\circ\text{C} = 297 \text{ K}$; $T_m = \frac{413 + 297}{2} = 355 \text{ K}$

$$Gr = \frac{g \beta \Delta T D^3}{\nu^2} = \frac{9.81 \times 1 \times (140 - 24) \times 0.6^3}{355 \times (21.46 \times 10^{-6})^2} = 1.50 \times 10^9$$

$$Pr = 0.699$$

$$Nu = 0.6 (Gr Pr)^{1/4}$$

$$\frac{hD}{k} = 0.6 (Gr Pr)^{0.25}$$

$$h = \frac{k \times 0.6}{D} (1.5 \times 10^9 \times 0.699)^{0.25} = \frac{30.38 \times 10^{-3} \times 0.6}{0.6} \times 180 = 5.47 \text{ W/m}^2\text{K}$$

$$Q_c = hA\Delta T = 5.47 \times 4\pi \times 0.3^2 \times (140 - 24) = 717.26 \text{ W}$$

Assuming bulb to be black body; $\epsilon = 1$

$$Q_r = \epsilon \sigma A_s (T_s^4 - T_\infty^4) = 1 \times 5.67 \times 10^{-8} \times 4\pi \times 0.3^2 \times (413^4 - 297^4) = 1366 \text{ W}$$

Total heat loss

$$Q = Q_c + Q_r = 717.26 + 1366 = 2083.29 \text{ W}$$

Power loss by natural convector

$$= \frac{717.26}{2083.29} \times 100 = 34.43 \% \quad \dots \text{Ans.}$$

Ex. 6.7.2: A vertical plate 2.2 m high and 1.4 m wide has been designed on free convection heating of a liquid. The temperature of the plate surface is maintained at 960°C while the temperature of liquid is 340°C. calculate the heat dissipation from both sides of the plate. For convection coefficient, use the co-relation.

$Nu = 0.13 [Gr Pr]^{0.333}$. The parameters of the liquid at 650°C are $\rho = 10^4 \text{ kg/m}^3$; $k = 13.02 \text{ W/m}\cdot\text{K}$; $C_p = 150.7 \text{ J/kg}\cdot\text{K}$; $\mu = 3.12 \text{ kg/m}\cdot\text{s}$.

Soln.:

Given: $H = 2.2 \text{ m}$; $W = 1.4 \text{ m}$
 $T_s = 960^\circ\text{C} = 960 + 273 = 1233 \text{ K}$
 $T_\infty = 340^\circ\text{C} = 340 + 273 = 613 \text{ K}$
 $Nu = 0.13 (Gr Pr)^{0.333}$

Properties of fluid at 650°C are:

$$\rho = 10^4 \text{ kg/m}^3$$

$$k = 13.02 \text{ W/m}\cdot\text{K}$$

$$C_p = 150.7 \text{ J/kg}\cdot\text{K}$$

$$\mu = 3.12 \text{ kg/m}\cdot\text{s} = 8.67 \times 10^{-4} \text{ kg/m}\cdot\text{s}$$

Heat dissipation,

$$Q = h \cdot 2A \cdot (T_s - T_\infty)$$

To find out h , we will use,

$$Nu = \frac{hL}{k} = 0.13 (Gr Pr)^{0.333} \quad (\text{given correlation})$$

$$Gr = \frac{g \beta \Delta T L^3}{\nu^2} = \frac{\rho^2 \cdot g \cdot \beta \cdot \Delta T \cdot L^3}{\mu^2} = \frac{(10^4)^2 \times 9.81 \times 1 \times (960 - 340) \times 2.2^3}{(650 + 273) \times (8.67 \times 10^{-4})^2} = \frac{10^8 \times 9.81 \times 620 \times 2.2^3}{923 \times (8.67 \times 10^{-4})^2} = 9.334 \times 10^{15}$$

$$Pr = \frac{\mu C_p}{k} = \frac{8.67 \times 10^{-4} \times 150.7}{13.02} = 0.01$$

Now, $Nu = \frac{hL}{k} = 0.13 (Gr Pr)^{0.333}$

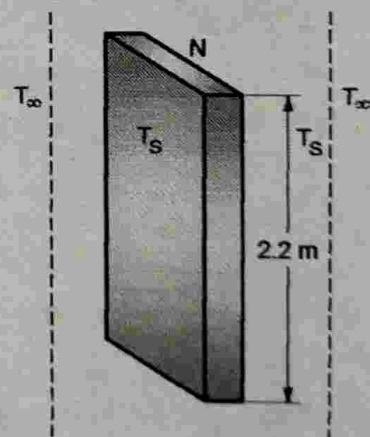


Fig. P. 6.7.2

$$\text{or } h = \frac{13.02 \times 0.13}{2.2} (9.334 \times 10^{15} \times 0.01)^{0.333} = 34,527.43 \text{ W/m}^2\text{K}$$

Hence total heat dissipation from both sides of plates

$$Q = h \times 2A (T_s - T_\infty) = 34527.43 \times 2 \times 2.2 \times 1.4 \times (96 - 340)$$



$$= 131867160.7 \text{ W}$$

$$= 131.8 \text{ MW}$$

...Ans.

Ex. 6.7.3 : A horizontal steam pipe of 100 mm OD runs through a room where the ambient air is at 20°C . If the outside surface of the pipe is at 180°C and the emissivity of the surface is 0.9, find out total heat loss per meter length of pipe. Properties of air at film temperature are $\rho = 0.946 \text{ kg/m}^3$, $\nu = 23.02 \times 10^{-6} \text{ m}^2/\text{s}$, $k = 0.03127 \text{ W/m}^\circ\text{C}$, $C_p = 1011.3 \text{ J/kg}^\circ\text{C}$, $\text{Pr} = 0.704$ and the correlation for the free convective heat loss is $\text{Nu} = 0.48 \text{ Ra}^{1/4}$.

Soln. :

Given :

$$\text{OD} = 100 \text{ mm} = 0.1 \text{ m};$$

$$T_\infty = 20^\circ\text{C}$$

$$T_s = 180^\circ\text{C} = 453 \text{ K};$$

$$\varepsilon = 0.9$$

$$\rho = 0.946 \text{ kg/m}^3;$$

$$\nu = 23.02 \times 10^{-6} \text{ m}^2/\text{s}$$

$$k = 0.03127 \text{ W/m}^\circ\text{C},$$

$$C_p = 1011.3 \text{ J/kg}^\circ\text{C},$$

$$\text{Pr} = 0.704;$$

$$\text{Nu} = 0.48 \text{ Ra}^{1/4}$$

Heat loss by radiation.

$$Q_r = \varepsilon \sigma A (T_s^4 - T_\infty^4)$$

$$= 0.9 \times 5.67 \times 10^{-8} \times \pi \times 0.1 \times 1 ((453)^4 - (293)^4)$$

$$= 556.67 \text{ W/m}$$

Heat loss by convection.

$$\beta = \frac{1}{100 + 273} = \frac{1}{373}$$

$$\text{Gr} = \frac{g \beta \Delta T D^3}{\nu^2} = \frac{9.81 \times 1 \times (180 - 20) \times 0.1^3}{373 \times (23.02 \times 10^{-6})^2}$$

$$= 7.94 \times 10^6$$

$$\text{Ra} = \text{Gr} \cdot \text{Pr} = 7.94 \times 10^6 \times 0.704 = 5590390.9$$

$$\text{Nu} = \frac{hD}{k} = 0.48 \text{ Ra}^{1/4}$$

$$h = \frac{0.03127 \times 0.48}{0.1} \times (5590390.9)^{1/4}$$

$$= 7.3 \text{ W/m}^2\text{K}$$

$$Q_c = h \cdot A \cdot \Delta T = 7.3 \times \pi \times 0.1 \times 1 \times (180 - 20)$$

$$= 366.67 \text{ W/m}$$

Total heat transfer rate.

$$Q = 556.67 + 366.67 = 923 \text{ W/m}$$

...Ans.

Convection Heat Transfer

Ex. 6.7.4 : A furnace 2 m wide $\times$ 1 m high and 3 m long having its outside temperature of 100°C while the surrounding air is at 30°C . Neglecting heat loss due to radiation and convection from base, find the rate of heat transfer from furnace surfaces. Properties of air are :

$$\rho = 1.02 \text{ kg/m}^3,$$

$$\nu = 18 \times 10^{-6} \text{ m}^2/\text{s},$$

$$C_p = 1005 \text{ J/kg K},$$

$$\mu = 20 \times 10^{-8} \text{ Ns/m},$$

$$k = 0.029 \text{ W/mK},$$

$$\text{Pr} = 0.693.$$

Soln. :

$$\text{Given : } B = 2 \text{ m}, \quad H = 1 \text{ m}, \quad L = 3 \text{ m},$$

$$T_s = 100^\circ\text{C}, \quad T_\infty = 30^\circ\text{C}.$$

$$\text{Find : } (a) \quad T_m \quad (b) \quad \beta = \frac{1}{T_m}$$

$$(c) \quad \text{Area of vertical surfaces and horizontal surface.}$$

$$(d) \quad \text{Gr} \cdot \text{Pr} \text{ and check for type of flow.}$$

$$(e) \quad \text{Using Nu relation, find } h.$$

$$(f) \quad Q.$$

Mean film temperature,

$$T_m = \frac{T_s + T_\infty}{2} = \frac{100 + 30}{2} = 65^\circ\text{C} = 338 \text{ K}.$$

$$\text{Coefficient of volumetric expansion, } \beta = \frac{1}{T_m} = \frac{1}{338} \text{ per K}$$

$$\Delta T = T_s - T_\infty = 100 - 30 = 70^\circ\text{C}$$

Area of vertical plates of the furnace,

$$A_1 = 2(B \times H + L \times H) = 2(2 \times 1 + 3 \times 1) = 10 \text{ m}^2$$

Area of horizontal plate with face up,

$$A_2 = B \times L = 2 \times 3 = 6 \text{ m}^2$$

(1) For vertical sides

$$\text{Gr} \cdot \text{Pr} = \frac{g \cdot \beta \cdot \Delta T L_c^3}{\nu^2} \times \text{Pr} \quad (\text{where, } L_c = H = 1 \text{ m})$$

$$= 9.81 \times \frac{1}{338} \times \frac{70 \times 1^3}{(18 \times 10^{-6})^2} \times 0.693$$

$$= 4.3455 \times 10^9$$

$$10^9 < \text{Gr} \cdot \text{Pr} < 10^{12}, \text{ the correlation for Nusselt Number}$$

is,

$$\text{Nu} = 0.13 (\text{Gr} \cdot \text{Pr})^{1/3}$$

$$= 0.13 \times (4.3455 \times 10^9)^{1/3} = 212.14$$

$$\text{But, } \text{Nu} = \frac{h \cdot L_c}{k};$$

$$h_1 = \frac{\text{Nu} \cdot k}{L_c} = \frac{212.14 \times 0.029}{1} = 6.152 \text{ W/m}^2\text{K}$$

Heat transfer rates from vertical sides,

$$Q_1 = h_1 \cdot A_1 \Delta T$$

$$= 6.152 \times 10 \times 70 = 4306.4 \text{ W}$$

2) Consider horizontal plate facing up

Characteristic length, $L_c = \frac{A}{P} = \frac{2 \times 3}{2 \times (2 + 3)} = 0.6$

$$Gr \cdot Pr = \frac{g \cdot \beta \cdot \Delta T \cdot L_c^3}{\nu^2} \cdot Pr$$

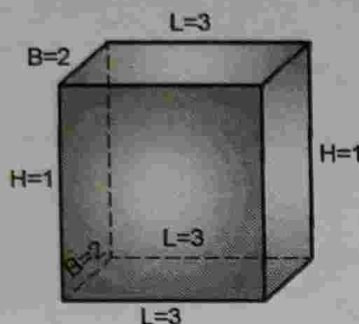


Fig. P. 6.7.4

$$Gr \cdot Pr = 9.81 \times \frac{1}{338} \times \frac{70 \times (0.6)^3}{(18 \times 10^{-6})^2} \times 0.693$$

$$= 9.3863 \times 10^8$$

Hence using $Nu = 0.15 (Gr \cdot Pr)^{1/3}$

$$= 0.15 (9.3863 \times 10^8)^{1/3}$$

$$= 147 \text{ W/m}^2 \text{ K}$$

$$Nu = \frac{h_2 \cdot L_c}{k}$$

$$h_2 = \frac{Nu \cdot k}{L_c} = \frac{147 \times 0.029}{0.6} = 7.105 \text{ W/m}^2 \text{ K}$$

Heat transfer rates from top surface,

$$Q_2 = h_2 \cdot A_2 (\Delta T)$$

$$= 7.105 \times 6 \times 70 = 2984.1 \text{ W}$$

(3) Total heat transfer rate from all sides of the furnace

$$Q = Q_1 + Q_2 = 4306.4 + 2984.1 \text{ W}$$

$$= 7290.5 \text{ W} \quad \dots \text{Ans.}$$

Ex. 6.7.5 : A circular disc insulated from other side of diameter 25 cm is exposed to air at 20°C. If the disc (Open surface) is maintained at 120°C, estimate heat transfer rate from it, when;

- Disc is kept horizontal with (open) hot surface facing upwards.
- Disc is kept horizontal with (open) hot surface facing downwards.
- Disc is kept vertical.

For air at 70°C, $k = 0.03 \text{ W/mK}$,

$$Pr = 0.697, \quad \nu = 2.076 \times 10^{-6}$$

Use the following correlations :

$$Nu = 0.14 (Gr \cdot Pr)^{0.334} \text{ for upward / top surface}$$

$$Nu = 0.27 (Gr \cdot Pr)^{0.25} \text{ for downward / bottom surface}$$

$$Nu = 0.59 (Gr \cdot Pr)^{0.25} \text{ for vertical surface.}$$

Soln. :

Given : $D = 25 \text{ cm} = 0.25 \text{ m}$, $T_\infty = 20^\circ\text{C}$,
 $T_s = 120^\circ\text{C}$.

Find

- T_m
- β
- L_c
- Gr
- $Gr \cdot Pr$ to check for application of Nu Equation.
- h
- Q .

Mean film temperature,

$$T_m = \frac{T_s + T_\infty}{2} = \frac{120 + 20}{2} = 70^\circ\text{C} = 343 \text{ K.}$$

Coefficient of volumetric expansion,

$$\beta = \frac{1}{T_m} = \frac{1}{343}$$

$$\Delta T = (T_s - T_\infty) = (120 - 20) = 100^\circ\text{C}$$

Characteristic length,

$$L = L_c = \frac{A}{P} = \frac{\left(\frac{\pi}{4}\right) D^2}{\pi D}$$

$$= \frac{D}{4} = \frac{0.25}{4} = 0.0625 \text{ m}$$

$$A = \frac{\pi}{4} D^2 = \frac{\pi}{4} (0.25)^2 = 0.04909 \text{ m}^2$$

$$Gr = \frac{g \cdot \beta \cdot \Delta T L^3}{\nu^2}$$

$$= 9.81 \times \frac{1}{343} \times \frac{100 \times (0.0625)^3}{(2.076 \times 10^{-6})^2}$$

$$= 1.6202 \times 10^8$$

$$Gr \cdot Pr = 1.6202 \times 10^8 \times 0.697 = 1.1293 \times 10^8$$

(1) Heat transfer rate when disc is facing up

$$Nu = 0.15 (Gr \cdot Pr)^{0.334} = 73.4$$

$$Nu = \frac{h \cdot L}{k}$$

$$h = \frac{Nu \cdot k}{L}$$

$$h = \frac{73.4 \times 0.03}{0.0625} = 35.23 \text{ W/m}^2 \text{ K}$$

$$Q = h \cdot A \cdot \Delta T = 35.23 \times 0.04909 \times 100$$

$$= 172.9 \text{ W} \quad \dots \text{Ans.}$$

(2) Heat transfer rate when disc is facing down

Since L_c remains the same,

$$\begin{aligned} \text{Nu} &= 0.27 (\text{Gr} \cdot \text{Pr})^{0.25} \\ &= 0.27 \times (1.1293 \times 10^8)^{0.25} = 27.833 \end{aligned}$$

$$h = \frac{\text{Nu} \cdot k}{L_c} = \frac{27.833 \times 0.03}{0.0625} = 13.36 \text{ W/m}^2 \text{ K}$$

$$Q = h \cdot A \cdot \Delta T = 13.36 \times 0.04909 \times 100$$

$$= 65.58 \text{ W} \quad \dots \text{Ans.}$$

(3) Heat transfer rate when disc is kept vertical

Characteristic length $L = L_c = D = 0.25 \text{ m}$ (for vertical plate)

$$\begin{aligned} \therefore \text{Gr} &= \frac{g \cdot \beta \cdot \Delta T \cdot L^3}{\nu^2} \\ &= 9.81 \times \frac{1}{343} \times \frac{100 \times (0.25)^3}{(2.07 \times 10^{-6})^2} \end{aligned}$$

$$= 104.3 \times 10^8$$

$$\begin{aligned} \text{Nu} &= 0.59 (\text{Gr} \cdot \text{Pr})^{0.25} \\ &= 0.59 (104.3 \times 10^8 \times 0.697)^{0.25} \\ &= 172.28 \end{aligned}$$

$$h = \frac{\text{Nu} \cdot k}{L} = \frac{172.28 \times 0.03}{0.25} = 20.673 \text{ W/m}^2 \text{ K}$$

$$Q = h \cdot A \cdot \Delta T = 20.673 \times 0.04909 \times 100$$

$$= 101.5 \text{ W} \quad \dots \text{Ans.}$$

Ex. 6.7.6 : Consider a thin square plate $0.6 \text{ m} \times 0.6 \text{ m}$ in dimensions, kept in a room at 30°C . One surface of the plate is maintained at 74°C while the other is insulated. Determine the rate of heat transfer from the plate if the plate is held vertical. What is the change in rate of heat transfer if the plate is kept horizontal with hot surface facing up.

Properties of air at mean film temperature :

$$\mu = 196.4 \times 10^{-7} \text{ Ns/m}^2, k = 28.15 \times 10^{-3} \text{ W/mK},$$

$$\text{Pr} = 0.7035, \quad \nu = 18.405 \times 10^{-6} \text{ m}^2/\text{s}.$$

Soln. :

Given : Area of plate, $A = 0.6 \times 0.6 = 0.36 \text{ m}^2$, $T_\infty = 30^\circ\text{C}$,

$$T_s = 74^\circ\text{C},$$

$$L = 0.6 \text{ m (vertical plate).}$$

$\therefore$ One side of plate is insulated, the heat will transfer only from one side of the plate. Since plate is kept in room in still air, the problem is considered of natural convection.

Mean film temperature,

$$\begin{aligned} T_m &= \frac{T_s + T_\infty}{2} = \frac{74 + 30}{2} \\ &= 52^\circ\text{C} = 325 \text{ K}. \end{aligned}$$

Coefficient of volumetric expansion,

$$\beta = \frac{1}{T_m} = \frac{1}{325}$$

Find

- Gr
- $\text{Ra} = \text{Gr} \cdot \text{Pr}$ for checking whether flow is laminar or turbulent
- Nu
- h using Nu
- Q .

Case I : When plate is held vertical

$$\text{Gr} = \frac{g \cdot \beta \cdot \Delta T \cdot L^3}{\nu^2} = g \times \frac{1}{T_m} \times \frac{(T_s - T_\infty) \times L^3}{\nu^2}$$

$$\begin{aligned} \text{Gr} &= 9.81 \times \frac{1}{325} \times \frac{(74 - 30) \times (0.6)^3}{(18.405 \times 10^{-6})^2} \\ &= 8.469 \times 10^8 \end{aligned}$$

$$\begin{aligned} \text{Ra} &= \text{Gr} \cdot \text{Pr} = (8.469 \times 10^8) \times 0.7035 \\ &= 5.958 \times 10^8 \end{aligned}$$

$\therefore 10^4 < \text{Ra} < 10^9$, Using $\text{Nu} = 0.59 \text{Ra}^{0.25}$

$$\begin{aligned} \text{Nu} &= 0.59 (\text{Ra})^{0.25} = 0.59 (5.958 \times 10^8)^{0.25} \\ &= 92.117 \end{aligned}$$

$$\text{Nu} = \frac{h \cdot L}{k}$$

$$\begin{aligned} h &= \frac{\text{Nu} \cdot k}{L} = \frac{92.117 \times (28.15 \times 10^{-3})}{0.6} \\ &= 4.325 \text{ W/m}^2 \text{ K} \end{aligned}$$

Heat transfer rate

$$\begin{aligned} Q &= h \cdot A \cdot (T_s - T_\infty) \\ &= 4.325 \times 0.36 \times (74 - 30) \\ &= 68.5 \text{ W} \end{aligned}$$

$\dots \text{Ans.}$

Case II : When plate is kept horizontal with hot surface on upper side

$$L = L_c = \frac{A}{P} = \frac{0.6 \times 0.6}{2(0.6 + 0.6)} = 0.15 \text{ m}$$

$$\begin{aligned} \text{Gr} \cdot \text{Pr} &= \frac{9.81 \times 1 \times (74 - 30) \times (0.15)^3}{325 \times (18.405 \times 10^{-6})^2} \times 0.7035 \\ &= 9.31 \times 10^4 \end{aligned}$$

Nusselt number is given by the relation

$$\begin{aligned} \text{Nu} &= 0.54 (\text{Gr} \cdot \text{Pr})^{0.25} \\ &= 0.54 (9.31 \times 10^4)^{0.25} = 29.83 \end{aligned}$$

$$\begin{aligned} \therefore h_1 &= \frac{\text{Nu} \cdot k}{L} = \frac{29.83 \times 0.02815}{0.15} \\ &= 5.6 \text{ W/m}^2 \text{ K} \end{aligned}$$



Heat transfer rate,

$$\begin{aligned} Q_1 &= h_1 \cdot A \times (T_s - T_\infty) \\ &= 5.6 \times 0.36 \times (74 - 30) \\ &= 88.7 \text{ W} \end{aligned}$$

...Ans.

Percentage change in rate of heat transfer

$$\begin{aligned} &= \frac{Q_1 - Q}{Q} \times 100 = \frac{(88.7 - 68.5)}{68.5} \times 100 \\ &= 29.5 \% \end{aligned}$$

...Ans.

Ex. 6.7.7 : A solid cylinder of steel of 12 cm diameter and 30 cm in length maintained at 380°C is suspended vertically in a large room at temperature 20°C. If the emissivity of the cylinder surface is 0.8, find total heat loss rate by the cylinder. Take properties of air at 200°C as follows :

$$\begin{aligned} C_p &= 1026 \text{ J/kg K}, \quad \rho = 0.746 \text{ kg/m}^3, \quad k = 0.0393 \text{ W/mK}, \\ \nu &= 34.85 \times 10^{-6} \text{ m}^2/\text{s}. \end{aligned}$$

Use the following correlations :

$$Nu = 0.56 (Gr \cdot Pr)^{0.25} \text{ for vertical surface}$$

$$Nu = 0.27 Ra^{0.25} \text{ for bottom horizontal surface}$$

$$Nu = 0.54 Ra^{0.25} \text{ for top horizontal surface.}$$

Soln. : This is the case of natural convection.

$$\begin{aligned} \text{Given : } D &= 12 \text{ cm} = 0.12 \text{ m}, L = 30 \text{ cm} = 0.3 \text{ m}, \\ T_s &= 380^\circ\text{C} = 653 \text{ K}, \\ T_\infty &= 20^\circ\text{C} = 293 \text{ K}, \quad \epsilon = 0.8. \end{aligned}$$

Find :

- T_m and check that properties of air given are in order.
- β
- Characteristic length for top and bottom horizontal surfaces vertical.
- $Gr \cdot Pr$ for each surface
- Nu
- h
- $Q_c + Q_r$

Mean film temperature,

$$T_m = \frac{T_s + T_\infty}{2} = \frac{380 + 20}{2} = 200^\circ\text{C}$$

Thus properties of air given are in order

Thermal coefficient of expansion,

$$\beta = \frac{1}{T_m} = \frac{1}{200 + 273} = 2.114 \times 10^{-3} \text{ K}^{-1}$$

At 200°C the given properties are,

$$\rho = 0.746 \text{ kg/m}^3, \quad C_p = 1026 \text{ J/kg K}$$

$$k = 0.0393 \text{ W/m K}, \quad \nu = 34.85 \times 10^{-6} \text{ m}^2/\text{s}$$

(i) For vertical surface

Characteristic length is, $L = 0.3 \text{ m}$

$$Gr \cdot Pr = \frac{g \cdot \beta \cdot \Delta T \cdot L^3}{\nu^2} \times Pr$$

$$\begin{aligned} Gr \cdot Pr &= \frac{9.81 \times (2.114 \times 10^{-3}) \times (380 - 20) (0.3)^3 \times 0.68}{(34.85 \times 10^{-6})^2} \\ &= 11.286 \times 10^7 \end{aligned}$$

For vertical cylinder, the given, correlations is,

$$Nu = 0.59 (Gr \cdot Pr)^{0.25}$$

$$= \frac{h \cdot L}{k}$$

$$\frac{h_1 \times 0.3}{0.0393} = 0.59 \times (11.286 \times 10^7)^{0.25}$$

$$\therefore h_1 = 7.97 \text{ W/m}^2 \text{ K}$$

Surface area,

$$\begin{aligned} A_1 &= \pi D L \\ &= \pi \times 0.12 \times 0.3 = 0.113 \text{ m}^2 \end{aligned}$$

(ii) For horizontal surface

Characteristic length,

$$L = \frac{A}{P} = \frac{\frac{\pi}{4} D^2}{\pi D} = \frac{D}{4} = 3 \text{ cm} = 0.03 \text{ m}$$

$$\text{Now, } Gr \cdot Pr = \frac{g \beta (\Delta T) L^3}{\nu^2} \times Pr$$

$$Gr \cdot Pr = \frac{9.81 \times (2.114 \times 10^{-3}) \times (380 - 20) (0.03)^3 \times 0.68}{(34.85 \times 10^{-6})^2}$$

$$Gr \cdot Pr = 1.1286 \times 10^5$$

For horizontal plate facing up (top surface of cylinder can be considered as a horizontal plate facing up) the correlation is,

$$\text{Given : } Nu = 0.54 \times (Gr \cdot Pr)^{0.25} = \frac{h_2 \cdot L_2}{k}$$

$$0.54 \times (1.1286 \times 10^5)^{0.25} = \frac{h_2 \times 0.3}{0.0393}$$

$$\therefore h_2 = 12.966 \text{ W/m}^2 \text{ K}$$

$$\text{Surface area, } A_2 = \frac{\pi}{4} D^2 = \frac{\pi}{4} \times (0.12)^2 = 0.01131 \text{ m}^2$$

(iii) For horizontal bottom surface the value of $Gr \times Pr$ will be same as previous case

$$\therefore Gr \cdot Pr = 1.1286 \times 10^5$$

And we can use the correlation,

$$\text{Given : } Nu = 0.27 (Gr \cdot Pr)^{0.25} = \frac{h_3 \cdot L_3}{k}$$

$$0.27 [(1.1286) \times 10^3]^{0.25} = \frac{h_3 \times 0.03}{0.0393}$$

$$h_3 = 6.483 \text{ W/m}^2 \text{ K}$$

Surface area,

$$A_3 = A_2 = 0.01131 \text{ m}^2$$

Total surface area,

$$\begin{aligned} A &= A_1 + A_2 + A_3 \\ &= 0.113 + 0.01131 + 0.01131 \\ &= 0.13562 \text{ m}^2 \end{aligned}$$

Total heat loss,

$$\begin{aligned} Q &= \text{Heat losses by convection from all surfaces} \\ &\quad + \text{Heat loss by radiation} \\ &= (h_1 A_1 + h_2 A_2 + h_3 A_3) (T_s - T_f) \\ &\quad + \epsilon \cdot \sigma \cdot A (T_s^4 - T_f^4) \\ &= (7.97 \times 0.113 + 12.966 \times 0.01131 + 6.483 \\ &\quad \times 0.01131) (380 - 20) + 0.8 \times (5.67 \times 10^{-8}) \\ &\quad \times 0.13562 (653^4 - 293^4) \\ &= 1476.6 \text{ W} \end{aligned}$$

...Ans.

Ex. 6.7.8 : A horizontal cylindrical rod of 4 cm diameter and 60 cm length is initially at a temperature of 124°C. Calculate the rate of heat lost from it, if it is exposed to still water at 30°C.

$$\begin{aligned} \text{Nu} &= 0.53 \text{ Ra}^{1/4} \text{ for } 10^4 < \text{Ra} < 10^9 \\ &= 0.13 \text{ Ra}^{1/3} \text{ for } 10^9 < \text{Ra} < 10^{12}. \end{aligned}$$

Properties of water :

$$C_p = 4191 \text{ J/kg K}, \quad k = 0.668 \text{ W/mK}, \quad \mu = 3.72 \times 10^{-4} \text{ kg/ms}$$

$$\rho = 937.7 \text{ kg/m}^3, \quad \beta = 6.286 \times 10^{-4} / \text{K}$$

Soln. :

$$\begin{aligned} \text{Given :} \quad D &= 4 \text{ cm} = 0.04 \text{ m}, \quad L = 60 \text{ cm} = 0.6 \text{ m}, \\ T_s &= 124^\circ\text{C}, \quad T_\infty = 30^\circ\text{C}. \end{aligned}$$

Find

- Gr Pr and product Gr Pr
- Compare Gr Pr with Ra choosing co-relation for Nu.
- Nu
- h
- Q

$$\Delta T = 124 - 30 = 94^\circ\text{C}$$

$\therefore$ Grashoff number

$$\begin{aligned} &= \frac{L^3 \cdot \rho^2 \cdot \beta^2 \cdot g \cdot \Delta T}{\mu^2} \\ &= \frac{(0.6)^3 \times (937.7)^2 \times (6.286 \times 10^{-4}) \times 9.81 \times 94}{(3.72 \times 10^{-4})^2} \\ &= 8.578 \times 10^{11} \end{aligned}$$

$$\text{Pr} = \frac{\mu C_p}{k} = \frac{(3.72 \times 10^{-4}) \times 4191}{0.668} = 2.334$$

$$\begin{aligned} \text{Ra} &= \text{Gr} \times \text{Pr} = (8.5763 \times 10^{11}) \times 2.334 \\ &= 2.0017 \times 10^{12} \end{aligned}$$

$$\begin{aligned} \therefore \text{Nu} &= 0.13 \times \text{Ra}^{1/3} = 0.13 (2.0017 \times 10^{12})^{1/3} \\ &= 1638.4 \end{aligned}$$

$$\text{Nu} = \frac{h \cdot L}{k}$$

$$1638.4 = \frac{h \times 0.6}{0.668}$$

$$\therefore h = 1824.1 \text{ W/m}^2 \text{ K}$$

Convective heat loss Q_c :

$$Q_c = hA \Delta T = h \times \pi \times d \times L \times \Delta T$$

$$Q_c = 1824.1 \times (\pi \times 0.04 \times 0.6) \times 94$$

$$Q_c = 12928.1 \text{ W}$$

...Ans.

Ex. 6.7.9 : A steel pipe of OD 0.15 m lies 2 m vertically and 8m horizontally in a large room with an ambient temperature of 30°C. The pipe surface is at 250°C and has an emissivity of 0.6. Estimate the total heat loss (due to convection and radiation) from the pipe to the atmosphere. Properties at film temperature : $\nu = 27.8 \times 10^{-6} \text{ m}^2/\text{s}$, $k = 0.035 \text{ W/mK}$, $\text{Pr} = 0.684$. [Take $\text{Nu} = 0.13 (\text{Gr.Pr})^{1/3}$ if the flow is turbulent OR $\text{Nu} = 0.53 (\text{Gr.Pr})^{1/4}$ if the flow is Laminar.]

Soln. :

$$\text{Outer diameter (OD)} = 0.15 \text{ m}$$

$$L = 2 \text{ m (Vertical)}$$

$$L = 8 \text{ m (horizontal)}$$

$$T_\infty = 30^\circ\text{C} ;$$

$$T_s = 250^\circ\text{C} ; \quad \epsilon = 0.6$$

Air properties

$$\nu = 27.8 \times 10^{-6} \text{ m}^2/\text{s} ;$$

$$k = 0.035 \text{ W/mK} ; \quad \text{Pr} = 0.684$$

Total Heat transfer will be from 2 m vertical pipe and 8 m horizontal pipe.

For vertical pipe : Heat flow by convection

Mean temperature,

$$T_m = \frac{250 + 30}{2} = 140^\circ\text{C} = 413 \text{ K}$$

$$\begin{aligned} \text{Gr} &= \frac{g \beta \Delta T L^3}{\nu^2} = \frac{9.81 \times 1 (250 - 30) \times 2^3}{413 \times (27.8 \times 10^{-6})^2} \\ &= 5.41 \times 10^{10} > 10^9 \text{ hence flow is turbulent} \end{aligned}$$



Therefore using,

$$Nu = 0.13(Gr Pr)^{1/4}$$

and $Nu = \frac{h \cdot L}{k}$

$$\frac{hL}{k} = 0.13 (5.41 \times 10^{10} \times 0.684)^{0.333}$$

$$h = \frac{0.035 \times 0.13 (5.41 \times 10^{10} \times 0.684)^{0.333}}{2}$$

$$= 7.52 \text{ W/m}^2\text{K}$$

$$Q_c = hA \Delta T$$

$$= 7.52 \times \pi \times 0.15 \times 2 \times (250 - 30)$$

$$= 1558.44 \text{ W}$$

For horizontal pipe, heat flow by convection

Characteristic length

$$= D = 0.15 \text{ m}$$

$$Gr = \frac{g \beta \Delta T D^3}{\nu^2}$$

$$= \frac{9.81 \times 1 \times (250 - 30) \times 0.15^3}{413 \times (27.8 \times 10^{-6})^2}$$

$$= 0.228 \times 10^8 < 10^9,$$

hence flow is laminar.

$$\text{Using } Nu = 0.53 (Gr Pr)^{1/4}$$

$$\frac{hD}{k} = 0.53 (0.228 \times 10^8 \times 0.684)^{0.25}$$

$$h = \frac{0.035 \times 0.53}{0.15} (0.228 \times 10^8 \times 0.684)^{0.25}$$

$$= 7.73 \text{ W/m}^2\text{K}$$

$$Q_r = h_r \cdot A \Delta T$$

$$= 7.73 \times \pi \times 0.15 \times 8 \times (250 - 30)$$

$$= 6407.86 \text{ W}$$

Heat transfer by radiation from entire pipe length

$$Q_r = \sigma \epsilon A (T_s^4 - T_\infty^4); \text{ where, } A = \pi DL$$

$$= (5.67 \times 10^{-8}) \times 0.6 \times (\pi \times 0.15 \times 10)$$

$$\times [(250 + 273)^4 - (30 + 273)^4]$$

$$= 10637.82 \text{ W}$$

Total heat loss from pipe

$$Q = Q_c + Q_r + Q_{rr}$$

$$= 1558.44 + 6407.86 + 10637.82$$

$$= 18604 \text{ W}$$

...Ans.

Ex. 6.7.10 : A furnace door, 1.5 m high and 1m wide is insulated from inside and has an outer surface temperature of 70°C. If the surrounding ambient air is at 30°C calculate steady state heat loss from the door. Take the properties from at film temperature 50°C are $\rho = 1.093 \text{ kg/m}^3$, $\nu = 17.95 \times 10^{-6} \text{ m}^2/\text{s}$, $Pr = 0.698$, $C_p = 1.005 \text{ kJ/kg} \cdot \text{K}$, use the correlation $Nu = 0.13 (Ra)^{1/3}$

Soln. :

$$A = \text{Height} \times \text{Width} = 1.5 \text{ m} \times 1 \text{ m} = 1.5 \text{ m}^2$$

$$T_s = 70^\circ\text{C}$$

$$T_\infty = 30^\circ\text{C}$$

$$T_m = \frac{70 + 30}{2} = 50^\circ\text{C} = 323 \text{ K};$$

$$\rho = 1.093 \text{ kg/m}^3,$$

$$\nu = 17.95 \times 10^{-6} \text{ m}^2/\text{s}, \quad Pr = 0.698,$$

$$C_p = 1.005 \text{ kJ/kg K}$$

$$Nu = 0.13 (Ra)^{1/3}$$

$$Gr = \frac{g \beta \Delta T L^3}{\nu^2} = \frac{9.81 \times 1 \times (70 - 30) \times 1.5^3}{323 \times (17.95 \times 10^{-6})^2}$$

$$= 1.2725 \times 10^{10}$$

$$Ra = Gr Pr = 1.2725 \times 10^{10} \times 0.698$$

$$= 8.88 \times 10^9$$

$$\therefore Nu = 0.13 Ra^{1/3}$$

$$= 0.13 (8.88 \times 10^9)^{0.333} = 267.2$$

$$Pr = \frac{\mu C_p}{k} = \frac{\rho \nu C_p}{k}$$

$$k = \frac{\rho \cdot \nu \cdot C_p}{Pr}$$

$$\therefore k = \frac{1.093 \times (17.95 \times 10^{-6}) \times 1.005 \times 10^3}{0.698}$$

$$= 0.028 \text{ W/mK}$$

$$\therefore \frac{h \cdot L}{k} = Nu; \quad h = \frac{k \cdot Nu}{L}$$

$$h = \frac{0.028 \times 267.2}{1.5} = 4.988 \text{ W/m}^2\text{K}$$

Heat transfer rate, Q

$$Q = h \cdot A \cdot \Delta T = 4.988 \times 1.5 \times 1 (70 - 30)$$

$$= 299.26 \text{ W}$$

...Ans.

Ex. 6.7.11 : A hot rectangular plate 5 cm $\times$ 3 cm maintained at 200°C is exposed to still air at 30°C. Calculate percentage increase in convective heat transfer rate if smaller side of the plate is held vertical than the bigger side. Neglect ITG of the thickness.

Use Correlation $Nu = 0.59 (Gr \cdot Pr)^{0.25}$

Air properties at 115°C : density = 0.91 kg/m³,

$C_p = 1.009 \text{ kJ/kg K}$,

$\mu = 22.65 \times 10^{-6} \text{ kg/ms}$, $k = 0.0331 \text{ W/mK}$.

Soln. :

Given : $T_s = 200^\circ\text{C}$, $T_\infty = 30^\circ\text{C}$.

Find : (a) T_m (b) β
(c) L (d) Gr

(e) $Gr \cdot Pr$ (f) Nu

(g) h (i) Q

$$T_m = \frac{T_s + T_\infty}{2} = \frac{200 + 30}{2} = 115^\circ\text{C} = 388 \text{ K}$$

$$\beta = \frac{1}{T_m} = \frac{1}{388}$$

Case I : Bigger side ($L = 5 \text{ cm}$) is held vertical

$$Gr = \frac{\rho^2 g \beta \Delta T L^3}{\mu^2} = \frac{0.91^2 \times 9.81 \times 1 \times (200 - 30) (0.05)^3}{388 (22.65 \times 10^{-6})^2} = 8.67 \times 10^5$$

$$Pr = \frac{\mu C_p}{k} = \frac{22.65 \times 10^{-6} \times 1009}{0.0331} = 0.69$$

$$Nu = 0.59 (Gr \cdot Pr)^{0.25} = 0.59 (8.67 \times 10^5 \times 0.69)^{0.25} = 16.41$$

$$Nu = \frac{h_L \cdot L}{k} = 16.41$$

$$h_L = 16.41 \times \frac{0.0331}{0.05} = 10.86 \text{ W/m}^2 \text{ K}$$

$$Q = hA \Delta T = 10.86 \times (0.05 \times 0.03 \times 2) (200 - 30) = 5.54 \text{ W} \quad \dots \text{Ans.}$$

Case II : When smaller side ($L = 3 \text{ cm}$) is held vertical

Since characteristic length has changed, Grashoff number will change, hence.

$$Gr = \frac{\rho^2 g \beta \Delta T L^3}{\mu^2} = \frac{0.91^2 \times 9.81 \times 1 \times (200 - 30) (0.03)^3}{388 (22.65 \times 10^{-6})^2} = 1.87 \times 10^5$$

$$Nu = 0.59 (Gr \cdot Pr)^{0.25} = 0.59 (1.87 \times 10^5 \times 0.69)^{0.25} = 11.18$$

$$h = \frac{Nu \cdot k}{L} = \frac{11.18 \times 0.0331}{0.03} = 12.33 \text{ W/m}^2 \text{ K}$$

$$Q = h_s A \Delta T = 12.33 \times (0.05 \times 0.03) \times 2 (200 - 30) = 6.288 \text{ W} \quad \dots \text{Ans.}$$

Increase in Heat Transfer Rate.

$$Q = \frac{6.288 - 5.54}{5.54} \times 100 = 13.5\% \quad \dots \text{Ans.}$$

Ex. 6.7.12 : The CPU of a personal computer has dimensions of $10 \text{ cm} \times 50 \text{ cm} \times 40 \text{ cm}$ height. Its surface temperature is 39°C . It is kept in still air at 15°C . Neglecting heat transfer from its bottom surface, find the rate of heat transfer from.

(i) all 4 vertical surface

(ii) its top surface

Properties at mean film temperature of 300 K are :

$$\rho = 1.1614 \text{ kg/m}^3, \quad C_p = 1007 \text{ J/kg K},$$

$$\nu = 15.89 \times 10^{-6} \text{ m}^2/\text{s},$$

$$k = 0.0263 \text{ W/mK}, \quad Pr = 0.707$$

Surface	Correlation	Range of 'Ra'
Vertical	$Nu = 0.59 Ra^{1/4}$ $= 0.1 Ra^{1/3}$	$10^4 - 10^9$ $10^9 - 10^{13}$
Horizontal	$Nu = 0.54 Ra^{1/4}$ $= 0.14 Ra^{1/3}$	$10^5 - 2 \times 10^7$ $2 \times 10^7 - 2 \times 10^{10}$

[Note : For a rectangular horizontal surface, characteristic dimension = Average of lengths of 2 sides of the rectangle]

Solution : Refer Fig. P. 6.7.12

Given : $T_s = 39^\circ\text{C}$, $T_\infty = 15^\circ\text{C}$.

Find

- T_m
- Gr , Pr and product of $Gr \cdot Pr$ for vertical faces and top surfaces.
- using Nu based on Ra , find h .
- Q .

$$\text{Mean film temperature, } T_m = \frac{39 + 15}{2} = 27^\circ\text{C} = 300 \text{ K}$$

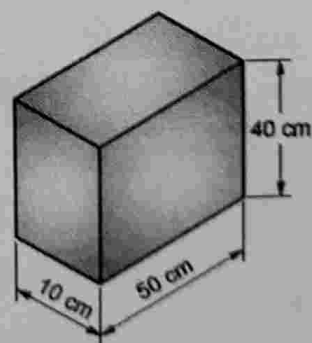


Fig. P. 6.7.12

Grashoff number,

$$Gr = \frac{g \beta \Delta T L^3}{\nu^2}$$

$$\text{where, } \Delta T = T_s - T_\infty = (39 - 15^\circ\text{C}) = 24^\circ\text{C}$$

(i) For vertical faces, $L = 0.4$ m

$$Gr = \frac{(9.81) \left(\frac{1}{300} \right) \times (24) (0.4)^3}{(15.89 \times 10^{-6})^2}$$

$$= 1.98925 \times 10^8$$

$$Ra = Gr \cdot Pr$$

$$Ra = (1.98925 \times 10^8) \times 0.707$$

$$= 1.4064 \times 10^8$$

$Ra < 10^9$, using

$$Nu = 0.59 \times Ra^{0.25} = 0.59 \times (1.4064 \times 10^8)^{0.25}$$

$$= 64.25$$

$$Nu = \frac{h \cdot L}{k}$$

$$64.25 = \frac{h \times 0.4}{0.0263}$$

$$h = 4.225 \text{ W/m}^2 \text{ K}$$

Total surface area of vertical faces,

$$A = 2 [0.4 \times 0.1 + 0.4 \times 0.5] = 0.48 \text{ m}^2$$

$$Q_{\text{vertical}} = h \cdot A \cdot \Delta T$$

$$= 4.225 \times 0.48 \times 24$$

$$= 48.67 \text{ W}$$

...Ans.

(ii) For horizontal top surface characteristic length

$$L = \frac{0.5 + 0.1}{2} = 0.3$$

$$Ra = Gr \cdot Pr = \frac{g \cdot \beta \cdot \Delta T \cdot L^3}{\nu^2} \times Pr$$

$$= 9.81 \times \left(\frac{1}{300} \right) \times \frac{24 \times 0.3^3}{(15.89 \times 10^{-6})^2} \times 0.707$$

$$= 59.33 \times 10^6$$

$$\therefore Nu = 0.14 \times Ra^{1/3} \text{ (given)}$$

$$= 0.14 \times (59.33 \times 10^6)^{1/3} = 54.6$$

$$h = \frac{Nu \cdot k}{L} = \frac{54.6 \times 0.0263}{0.3}$$

$$= 4.787 \text{ W/m}^2 \text{ K}$$

$$Q_{\text{horizontal}} = h \cdot A \cdot \Delta T = 4.787 \times (0.1 \times 0.5) \times 24$$

$$= 5.74 \text{ W}$$

...Ans.

Ex. 6.7.13 : A hot plate $1 \text{ m} \times 0.5 \text{ m}$ at 180°C is kept in still air at 20°C with 0.5 m side vertical. The plate has a mass of 20 kg and specific heat of 0.4 kJ/kg K . If convection takes place from both the sides of the plate, determine heat transfer coefficient, initial rate of cooling the plate and time required in cooling of the plate from 120°C to 80°C . Use the following thermo physical properties of the air.

$$k = 0.032 \text{ W/mK}, \quad \nu = 23.13 \times 10^{-6} \text{ m}^2/\text{s}, \quad Pr = 0.688$$

$$Nu = 0.14 Ra^{0.33} \text{ if turbulent } Ra > 10^5$$

$$Nu = 0.59 Ra^{0.25} \text{ if laminar } Ra < 10^5$$

Soln. : Refer Fig. P. 6.7.13.

Given : For plate : $C_p = 400 \text{ J/kg K}$,

$$m = 20 \text{ kg}, \quad T_s = 180^\circ\text{C},$$

$$T_\infty = 20^\circ\text{C} \quad T = 80^\circ\text{C}.$$

Find

- Gr and $Ra = Gr \cdot Pr$ to check for type of flow.
- Use appropriate co-relation for Nu and find h .
- Q
- Rate of cooling
- Time for cooling from T_s to T .

$$T_w = \frac{T_s + T_\infty}{2} = \frac{180 + 20}{2}$$

$$= 100^\circ\text{C} = 373 \text{ K}$$

$$\Delta T = T_s - T_\infty = 180 - 20 = 160^\circ\text{C}$$

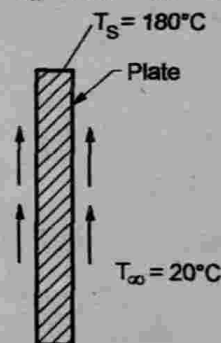


Fig. P. 6.7.13

$$Gr = \frac{L^3 g \beta \Delta T}{\nu^2}$$

$$Gr = \frac{0.5^3 \times 9.81 \times \left(\frac{1}{273 + 100} \right) \times (180 - 20)}{(23.13 \times 10^{-6})^2}$$

$$Gr = 982.191 \times 10^6$$

$$Ra = Gr \cdot Pr = 982.191 \times 10^6 \times 0.688$$

$$= 675.747 \times 10^6 > 10^5$$

(1) Heat transfer coefficient, h

$$Nu = 0.14 Ra^{0.33} = 114.8$$

$$Nu = 0.14 \times [675.747 \times 10^6]^{0.33}$$

$$Nu = \frac{h \cdot L}{k}$$

$$114.8 = \frac{h \times 0.5}{0.032}$$

$$h = 7.3474 \text{ W/m}^2 \text{ K}$$

...Ans.

(2) Initial rate of cooling of plate °C/min

Heat lost from both sides,

$$\begin{aligned} Q &= 2 [h A (T_s - T_\infty)] \\ &= 2 \times [7.3474 \times (1 \times 0.5) \times (180 - 20)] \\ &= 1175.584 \text{ W} \end{aligned}$$

But Rate heat lost = Rate of cooling

$$\text{i.e. } Q = m \cdot C_p \left(\frac{dT}{dt} \right)$$

$$1175.584 = 20 \times 400 \times \frac{dT}{dt}$$

Rate of cooling,

$$\begin{aligned} \frac{dT}{dt} &= \frac{1175.584}{20 \times 400} \\ &= 0.1469 \text{ °C/s or } 8.816 \text{ °C/min} \quad \dots \text{Ans.} \end{aligned}$$

(3) Time required in cooling the plate from 120°C to 80°C. i.e. $T_s = 120^\circ\text{C}$, $T = 80^\circ\text{C}$.

Instantaneous heat lost by the plate

$$\begin{aligned} 2 h A (T_s - T_\infty) &= -m C_p \frac{dT}{dt} \\ \frac{-m C_p}{2 h A} \int_{T_s}^T \frac{dT}{T - T_\infty} &= \int_0^t dt \end{aligned}$$

(a) Time required to cool upto 120°C

$$\begin{aligned} t &= \frac{-m C_p}{2 h A} \ln \left[\frac{T - T_\infty}{T_s - T_\infty} \right] \\ &= \frac{-20 \times 400}{2 \times 7.3474 \times 0.5} \ln \left[\frac{120 - 20}{180 - 20} \right] \\ &= \frac{-20 \times 400}{2 \times 7.3474 \times 0.5} \ln \left[\frac{100}{160} \right] \\ t &= 511.74 \text{ sec.} \quad \dots \text{Ans.} \end{aligned}$$

(b) Similarly time required for cooling to 80°C.

$$\begin{aligned} t_1 &= \frac{-20 \times 400}{2 \times 7.3474 \times 0.5} \ln \left[\frac{80 - 20}{180 - 20} \right] \\ t_1 &= 1067.94 \text{ sec.} \quad \dots \text{Ans.} \end{aligned}$$

Time required to cool from 120°C to 80°C

$$\begin{aligned} &= 1067.94 - 511.74 \\ &= 556.2 \text{ sec.} \quad \dots \text{Ans.} \end{aligned}$$

Ex. 6.7.14 : A solid cylinder of steel (density = 8000 kg/m³, $C_p = 0.42 \text{ kJ/kg}$) of 12 cm diameter and 30 cm length at 380°C is suspended vertically in a large room at temperature 20°C. If the emissivity of cylinder surface is 0.8, find total heat loss rate by the cylinder and initial rate of cooling.

Take properties of air at 200°C as follows :

$$\begin{aligned} C_p &= 1026 \text{ J/kg K}, \quad \rho = 0.746 \text{ kg/m}^3, \quad k = 0.0393 \text{ W/mK}, \\ \nu &= 34.85 \times 10^{-6} \text{ m}^2/\text{s} \end{aligned}$$

Use the following correlations :

$$Nu = 0.56 (Gr \cdot Pr)^{0.25} \text{ for vertical surface}$$

$$Nu = 0.27 (Ra)^{0.25} \text{ for lower horizontal surface}$$

$$Nu = 0.54 (Ra)^{0.25} \text{ for upper horizontal surface}$$

Soln. :

$$\begin{aligned} \text{Given : } D &= 12 \text{ cm}, \quad L = 30 \text{ cm} = 0.3 \text{ m}, \quad T_\infty = 20^\circ\text{C}, \\ \epsilon &= 0.8, \quad T_s = 200^\circ\text{C} = 473 \text{ K}, \\ T_m &= \frac{T_s + T_\infty}{2}, \quad T_m = \frac{380 + 20}{2} = 200^\circ\text{C}, \\ \beta &= \frac{1}{T_m} = \frac{1}{473}, \quad \Delta T = (380 - 20)^\circ\text{C}. \end{aligned}$$

Refer Fig. P. 6.7.14.

Line of Approach

We have to find out heat flow rate $Q = ?$

Heat flow will take place by convection and radiation.

$$\text{Radiant heat flow } Q_r = \epsilon_s \sigma A_s (T_s^4 - T_\infty^4)$$

$$\text{Heat flow by convection } Q_c = h A \Delta T$$

Since h will be different for different surfaces i.e. h_t for top, h_b for bottom and h_v for vertical surfaces, we should first find out h_t , h_b and h_v by given Nu co-relations.

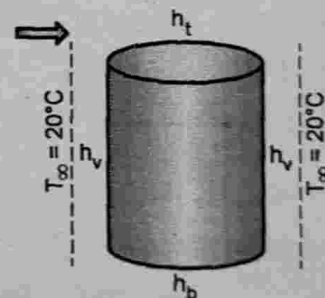


Fig. P. 6.7.14

We can now find out Q_c for different surfaces. Add up all

Q_r and Q_c to get total heat flow rate Q .

(1) For Vertical Surface

$$\begin{aligned} Gr &= \frac{g \beta \Delta T L^3}{\nu^2} = \frac{9.81 \times (380 - 20) \times 0.3^3}{(34.85 \times 10^{-6})^2} \\ &= 1.66 \times 10^8 \end{aligned}$$

$$Pr = \frac{\mu C_p}{k} = \frac{\rho \nu C_p}{k}$$



$$\begin{aligned}
 &= \frac{0.746 \times 34.85 \times 10^{-6} \times 1026}{0.0393} \\
 &= 0.679 \\
 \text{Nu} &= 0.56 (\text{Gr Pr})^{0.25} \\
 &= 0.56 (1.66 \times 10^5 \times 0.679)^{0.25} \\
 &= 57.69 \\
 h_c &= \frac{k \cdot \text{Nu}}{L} \\
 &= \frac{0.0393 \times 57.69}{0.03} = 7.56 \text{ W/m}^2 \text{ K}
 \end{aligned}$$

(2) For Top Horizontal Surface

Characteristic length,

$$\begin{aligned}
 L &= \frac{A}{P} = \frac{\pi D^2}{4\pi D} = \frac{D}{4} \\
 &= \frac{12}{4} = 3 \text{ cm} = 0.03 \text{ m}
 \end{aligned}$$

$$\begin{aligned}
 \text{Gr} &= \frac{g \beta \Delta T L^3}{\nu^2} \\
 &= \frac{9.81 \times (380 - 20) \times 0.03^3}{473 \times (34.85 \times 10^{-6})^2}
 \end{aligned}$$

$$\text{Gr} = 1.66 \times 10^5$$

$$\text{Pr} = \frac{\mu C_p}{k} \text{ remains the same}$$

$$\text{i.e. } \text{Pr} = 0.679$$

$$\begin{aligned}
 \text{Nu} &= 0.54 (\text{Gr Pr})^{0.25} \\
 &= 0.54 (1.66 \times 10^5 \times 0.679)^{0.25} = 9.89
 \end{aligned}$$

$$\begin{aligned}
 h_c &= \frac{\text{Nu} \cdot k}{L} = \frac{9.89 \times 0.0393}{0.03} \\
 &= 12.96 \text{ W/m}^2 \text{ K}
 \end{aligned}$$

(3) For Bottom Horizontal Surface

$$\begin{aligned}
 \text{Characteristic length } L &= \frac{A}{P} = \frac{\pi D^2}{4\pi D} = \frac{D}{4} \\
 &= \frac{12}{4} = 3 \text{ cm} = 0.03 \text{ m}
 \end{aligned}$$

$$\begin{aligned}
 \text{Gr} &= \frac{g \beta \Delta T L^3}{\nu^2} = \frac{9.81 \times (380 - 20) \times 0.03^3}{473 \times (34.85 \times 10^{-6})^2} \\
 &= 1.66 \times 10^5
 \end{aligned}$$

$$\text{Pr} = \frac{\mu C_p}{k} = 0.679 \text{ (same as above)}$$

$$\text{Nu} = 0.27 (\text{Gr Pr})^{0.25}$$

$$\text{Nu} = 0.27 (1.66 \times 10^5 \times 0.679)^{0.25} = 4.95$$

$$h_b = \frac{\text{Nu} \cdot k}{L} = \frac{4.95 \times 0.0393}{0.03}$$

$$h_b = 6.48 \text{ W/m}^2 \text{ K}$$

Hence total heat flow by convection

$$\begin{aligned}
 Q_c &= h_c \pi D L (T_s - T_\infty) + h_b \cdot \frac{\pi}{4} D^2 \cdot (T_s - T_\infty) \\
 &\quad + h_b \frac{\pi}{4} D^2 (T_s - T_\infty) \\
 &= 7.56 \times \pi \times 0.12 \times 0.3 (380 - 20) \\
 &\quad + 12.96 \times \frac{\pi}{4} \cdot 0.12^2 (380 - 20) \\
 &\quad + 6.48 \times \frac{\pi}{4} \cdot 0.12^2 (380 - 20) = 386.76 \text{ W}
 \end{aligned}$$

Heat loss by Radiation

$$\begin{aligned}
 Q_r &= \epsilon_1 \sigma A_1 \\
 &= \epsilon_1 \cdot \sigma \left(\pi D L + 2 \times \frac{\pi}{4} D^2 \right) (T_s^4 - T_\infty^4)
 \end{aligned}$$

$$\begin{aligned}
 Q &= 0.8 \times 5.67 \times 10^{-8} \times \left(\pi \times 0.12 \times 0.3 + 2 \times \frac{\pi}{4} \times 0.12^2 \right) \\
 &\quad (653^4 - 293^4) \\
 &= 1073.96 \text{ W}
 \end{aligned}$$

Hence total heat flow by convection and Radiation,

$$Q = Q_c + Q_r = 386.76 + 1073.96 = 1460.72 \text{ W}$$

To obtain Initial Rate of Cooling $Q = -m C_p \frac{dT}{dt}$

$$m = \rho V = \frac{8000 \times \pi (0.12)^2 \times 0.3}{4} = 27.13 \text{ kg}$$

$$\begin{aligned}
 \therefore \frac{dT}{dt} &= \frac{1460.72}{420 \times 27.13} \\
 &= 0.128^\circ \text{C/sec} = 7.69^\circ \text{C/min} \quad \dots \text{Ans.}
 \end{aligned}$$

Ex. 6.7.15 : A horizontal flat circular metallic plate is kept on its flat surface on a terrace ground in sunlight. Radius of plate = 2 m. It attains a steady state temperature of 77°C . The ambient temperature is 27°C . Calculate the rate at which solar heat energy is received by the plate by means of convection only. Also calculate the rate of heat transfer per unit area of the plate. Take characteristic length of plate = $\frac{A_{\text{surface}}}{\text{Perimeter}}$

For horizontal plate with hot surface up.

$$\text{Nu} = 0.13 (\text{Gr Pr})^{1/3} \text{ for } \text{Gr Pr} < 2 \times 10^8$$

$$= 0.16 (\text{Gr Pr})^{1/3} \text{ for } \text{Gr Pr} > 2 \times 10^8$$

Use the following properties of air :

$$k = 0.0282 \text{ W/mK}, \quad \nu = 18.23 \times 10^{-6} \text{ m}^2/\text{s}, \quad \text{Pr} = 0.7025$$

Soln. :

Given : $R = 2 \text{ m}$, $T_s = 77^\circ\text{C}$, $T_\infty = 27^\circ\text{C}$,
Mean film temperature, $T_m = \frac{T_s + T_\infty}{2} = \frac{77 + 27}{2} = 52^\circ\text{C}$.

Find

- (a) Thermal coefficient of expansion, β
(b) Characteristic length L_c ,
(c) Check for Gr Pr hence find Nu .
(d) h (e) Q (f) Q/A

Thermal coefficient of expansion,

$$\beta = \frac{1}{T_m} = \frac{1}{52 + 273} = \frac{1}{325} \\ = 3.077 \times 10^{-3} \text{ K}^{-1}$$

Characteristic length,

$$L = \frac{\text{Surface area}}{\text{Perimeter}} = \frac{\pi R^2}{2\pi R} = \frac{R}{2} = \frac{2}{2} = 1 \text{ m}$$

$$\text{Gr} \cdot \text{Pr} = \left[\frac{g \cdot \beta \cdot \Delta T \cdot L^3}{\nu^2} \right] \text{Pr} \\ = \left[\frac{9.81 \times (3.077 \times 10^{-3}) (77 - 27) 1^3}{(18.23 \times 10^{-6})^2} \right]$$

$$\times 0.7025$$

$$= 31.903 \times 10^6$$

$\text{Gr} \cdot \text{Pr} > 2 \times 10^6$, thus using the relation for Nu as :

$$Nu = 0.16 (\text{Gr} \cdot \text{Pr})^{1/3} \\ = 0.16 (31.903 \times 10^6)^{1/3} \\ = 235.54$$

$$\text{But, } Nu = \frac{h \cdot L}{k};$$

$$h = \frac{Nu \cdot k}{L_c} = \frac{235.54 \times 0.0282}{1}$$

$$h = 6.642 \text{ W/m}^2 \text{ K}$$

Rate of heat energy received by convection, ($A = \pi R^2$)

$$Q = h \cdot A \cdot (T_s - T_\infty) \\ = 6.642 \times (\pi \times 2^2) \times (77 - 27) \\ = 4173.29 \text{ W} \quad \dots \text{Ans.}$$

Rate of heat transfer per unit area

$$= \frac{Q}{A} = \frac{4173.25}{\pi \times (2)^2} = 332.1 \text{ W/m}^2 \quad \dots \text{Ans.}$$

Ex. 6.7.16 : A thin hot plate 0.5 m high and 2 m long is kept at 200°C having the surrounding air at 20°C . The mass of the plate is 40 kg. The heat is convected naturally from both sides. Find :

- (a) Initial rate of cooling of plate.

- (b) Time required to cool the plate from 150°C to 50°C .

Assume thermophysical properties of air at mean temperature as :

$$k = 0.032 \text{ W/mK}, \quad \nu = 23.5 \times 10^{-6} \text{ m}^2/\text{s}, \quad \text{Pr} = 0.7$$

$$\text{Specific heat of plate, } C = 380 \text{ J/kg K.}$$

Soln. :

Given : $L_c = H = 0.5 \text{ m}$, $L = 2 \text{ m}$, $T_s = 200^\circ\text{C}$,
 $T_\infty = 20^\circ\text{C}$, $m = 40 \text{ kg}$.

Mean film temperature,

$$T_m = \frac{T_s + T_\infty}{2} = \frac{200 + 20}{2} \\ = 110^\circ\text{C} = 383 \text{ K}$$

$\therefore$ Coefficient of volumetric expansion,

$$\beta = \frac{1}{T_m} = \frac{1}{383}$$

$$\Delta T = T_s - T_\infty = 200 - 20 = 180^\circ\text{C}.$$

Since heat is convected from both sides,

Surface area,

$$A = 2 \times (H \times L) = 2 \times (0.5 \times 2) = 2 \text{ m}^2.$$

$$\text{Gr} \cdot \text{Pr} = \frac{g \cdot \beta \cdot \Delta T \cdot L^3}{\nu^2} \times \text{Pr} \\ = 9.81 \times \frac{1}{383} \times \frac{(180) \times 0.5^3}{(23.5 \times 10^{-6})^2} \times 0.7 \\ = 7.3 \times 10^6$$

$10^4 < \text{Gr} \cdot \text{Pr} < 7.3 \times 10^6$, flow is laminar.

$$Nu = 0.59 (\text{Gr} \cdot \text{Pr})^{0.25} = 96.98 = \frac{h \cdot L_c}{k}$$

$$h = \frac{Nu \cdot k}{L_c} = \frac{96.98 \times 0.032}{0.5} = 6.207 \text{ W/m}^2 \text{ K}$$

- (i) Initial rate of cooling $\left(\frac{dT}{dt}\right)$, from law of conservation of energy,

Rate of decrease of internal energy = Rate of heat convection from plate.

$$m \cdot C \frac{dT}{dt} = h \cdot A \cdot (T_s - T_\infty)$$

$$400 \times 380 \times \frac{dT}{dt} = 6.207 \times 2 \times (200 - 20)$$

$$\frac{dT}{dt} = 0.0147^\circ\text{C} / \text{second} \quad \dots \text{Ans.}$$



- (ii) Time required to cool the plate from $T_1 = 150^\circ\text{C}$ to $T_2 = 50^\circ\text{C}$

$$\frac{T_2 - T_\infty}{T_1 - T_\infty} = e^{-\frac{h \cdot A}{\rho V C} \cdot t} = e^{-\frac{h \cdot A}{mC} \cdot t}$$

$$\therefore t = -\frac{m \cdot C}{h \cdot A} \ln \left(\frac{T_2 - T_\infty}{T_1 - T_\infty} \right)$$

$$t = -\frac{400 \times 380}{6.207 \times 2} \ln \left(\frac{50 - 20}{150 - 20} \right)$$

$$= 17954.2 \text{ seconds}$$

$$= \frac{17954.2}{60} = 299.236 \text{ min} \quad \dots \text{Ans.}$$

Ex. 6.7.17 : An incandescent lamp is approximated as a 75 mm diameter sphere. The temperature of the glass bulb of a 60 W lamp is 110°C when the ambient air is at 20°C . Estimate the heat transfer rate from the bulb surface by natural convection and by radiation if emissivity of the glass is 0.7. Properties of air at 65°C are $\mu = 2.03 \times 10^{-5} \text{ Ns/m}^2$, $\rho = 1.044 \text{ kg/m}^3$, $C_p = 1007 \text{ J/kg}\cdot\text{K}$, $k = 0.02845 \text{ W/m}\cdot\text{K}$, $\text{Pr} = 0.7188$.

Use the correlation $\text{Nu} = 0.589 \text{ Ra}^{1/4}$ and $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4$.

Soln. :

$$D = 75 \text{ mm} = 75 \times 10^{-3} \text{ m}$$

$$\therefore R = \frac{D}{2} = \frac{75 \times 10^{-3}}{2} = 37.5 \times 10^{-3} \text{ m}$$

$$T_1 = 110^\circ\text{C} = 110 + 273 = 383 \text{ K}$$

$$T_\infty = 20^\circ\text{C} = 293 \text{ K}$$

$$\epsilon = 0.7$$

$\therefore$ Mean film temperature, T_m :

$$T_m = \frac{T_1 + T_\infty}{2}$$

$$T_m = \frac{110 + 20}{2} = 65^\circ\text{C}$$

$$= 65 + 273 \text{ K} = 338 \text{ K}$$

$\therefore$ Coefficient of volumetric expansion, β :

$$\beta = \frac{1}{T_m} = \frac{1}{338} = 2.95857 \times 10^{-3} \text{ K}^{-1}$$

$$\nu = \frac{\mu}{\rho} = \frac{2.03 \times 10^{-5}}{1.044}$$

$$= 1.9444 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Gr} = \frac{g\beta\Delta T L^3}{\nu^2} = \frac{g\beta\Delta T D^3}{\nu^2} \quad (\because L = D)$$

$$\therefore \text{Gr} = \frac{9.81 \times 2.95857 \times 10^{-3} \times (110 - 20) \times (75 \times 10^{-3})^3}{(1.9444 \times 10^{-5})^2}$$

$$\therefore \text{Gr} = 2.9147 \times 10^6$$

$$\text{Ra} = \text{Gr Pr}$$

$$= 2.9147 \times 10^6 \times 0.7188$$

$$= 2.0951 \times 10^6$$

$$\text{Nu} = 0.589 (\text{Ra})^{1/4}$$

$$\therefore \frac{hL}{k} = 0.589 (2.0951 \times 10^6)^{1/4}$$

$$\therefore \frac{hD}{k} = 0.589 (2.0951 \times 10^6)^{1/4}$$

$$\therefore h = \frac{k}{D} \times 0.589 \times (2.0951 \times 10^6)^{1/4}$$

$$\therefore h = \frac{0.02845}{0.075} \times 0.589 \times (2.0951 \times 10^6)^{1/4}$$

$$\therefore h = 8.5 \text{ W/m}^2\cdot\text{K}$$

Heat transfer rate, Q

Heat transfer rate by convection,

$$Q_c = hA\Delta T$$

$$= 8.5 \times 4\pi \times (37.5 \times 10^{-3})^2 \times (110 - 20)$$

$$= 8.5 \times 4\pi \times 0.0375^2 \times (110 - 20)$$

$$Q_c = 13.518 \text{ W}$$

Heat transfer rate by radiation,

$$Q_r = \epsilon \sigma A (T_1^4 - T_\infty^4)$$

$$= 0.7 \times 5.67 \times 10^{-8} \times 4\pi (0.0375)^2 \times [383^4 - 293^4]$$

$$= 9.922 \text{ W}$$

$$\therefore Q = Q_c + Q_r = 13.518 + 9.922$$

$$\therefore Q = 23.44 \text{ W}$$

...Ans.

□□□